

# $q$ -Binomial Coefficient Calculus

**Gaussian Kernels, Bell Polynomials,  
Basis Changes, Parameter Jets, and Inversion**

*An extension of **Combinatorial Coefficient Calculus***

A self-contained development with detailed proofs

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# Abstract

This article extends the basis-change and coefficient-extraction framework of *Combinatorial Coefficient Calculus* to Gaussian, or  $q$ -binomial, coefficients. The extension is organized around mathematical operations, rather than a list of disconnected identities. Finite products produce Gaussian kernels; logarithms of those products produce power sums; Bell polynomials recover the coefficients; Newton interpolation produces  $q$ -Stirling arrays; and ordinary formal composition and Lagrange inversion transport the resulting formulae to implicit series.

The development includes Gaussian convolution and inversion, finite-field linearization, terminating  $q$ -Chu–Vandermonde sums, principal specializations of symmetric functions, a master coefficient formula for arbitrary powers of finite geometric products, partial-fraction proofs of divisor-type identities, Jackson differentiation and geometric interpolation, two explicitly distinguished  $q$ -Stirling normalizations, and a weighted Bell–Riordan calculus. It also provides recurrence-free coefficients of Gaussian polynomials through divisor sums, all-order expansions at  $q = 1$ , and a uniform deflation procedure giving all Taylor coefficients at every root of unity.

The last analytic application constructs Gaussian extrapolation filters. Their complete error response and exact noise norm are computed, and repeated spectral roots extend the construction to polynomial–logarithmic errors. Every theorem used in the development is proved or reduced explicitly to a proved result. Classical identities are identified as such; the synthesis and further consequences are not presented as claims of historical priority. A companion standard-library Python program performs exact, deterministic normalization checks. These checks supplement, rather than replace, the proofs.

# Relation to the source article

The source article [1] develops two central mechanisms: changing to a basis adapted to an operator, and using Bell polynomials to recover coefficients from component data. It also separates formal statements from analytic statements and keeps Bernoulli numbers distinct from Bell polynomials. This extension retains that organization and those distinctions.

Its first point of attachment is the source’s Newton expansion: the additive nodes  $0, 1, 2, \dots$  are replaced either by geometric nodes  $a, aq, aq^2, \dots$  or by  $q$ -integer nodes  $[0]_q, [1]_q, [2]_q, \dots$ . These are related but not identical constructions. Its second point of attachment is the exponential formula: a finite geometric alphabet has power sums  $[N]_{qr}$ , so the same ordinary Bell polynomials already used in the source immediately encode  $q$ -binomial identities. Its third point of attachment is ordinary composition and reversion. A change to  $q$ -factorial coefficient normalization changes the matrices, not the meaning of ordinary substitution.

This is a standalone mathematical extension, not a replacement of the canonical article and not a claim that its additional results have been formalized in Lean. All notation required for compilation is defined in this file; the repository’s shared notation file is not needed. The typography follows the source’s Libertinus, olive, and coral conventions.

## What “an extension” means here

The finite  $q$ -binomial theorem, Gaussian inversion,  $q$ -Chu–Vandermonde, Carlitz-type arrays, divisor identities, and  $q$ -Lucas are established mathematics. They are rederived to connect their hypotheses and normalizations. The main additional value is the common coefficient machinery, including explicit all-order root-of-unity jets and full geometric-filter error formulae. No exhaustive novelty claim is made.

# Contents

|                                                                                     |           |
|-------------------------------------------------------------------------------------|-----------|
| <b>Abstract</b>                                                                     | <b>i</b>  |
| <b>Relation to the source article</b>                                               | <b>ii</b> |
| <b>1 The coefficient-calculus setting</b>                                           | <b>1</b>  |
| 1.1 Rings, indices, and specialization . . . . .                                    | 1         |
| 1.2 Bell and Bernoulli normalizations . . . . .                                     | 1         |
| 1.3 Three distinct operations . . . . .                                             | 2         |
| <b>2 Gaussian polynomials and finite products</b>                                   | <b>3</b>  |
| 2.1 A division-free definition . . . . .                                            | 3         |
| 2.2 The finite and reciprocal binomial theorems . . . . .                           | 4         |
| 2.3 A formal infinite-binomial identity without a topology error . . . . .          | 4         |
| <b>3 Convolution, inversion, and linearization</b>                                  | <b>6</b>  |
| 3.1 Two-block and many-block Vandermonde identities . . . . .                       | 6         |
| 3.2 Gaussian inversion and the subspace lattice . . . . .                           | 6         |
| 3.3 A positive linearization formula . . . . .                                      | 7         |
| 3.4 Pochhammer convolution and terminating hypergeometric sums . . . . .            | 7         |
| <b>4 Symmetric functions and the Bell master formula</b>                            | <b>9</b>  |
| 4.1 Elementary, complete, and power-sum coordinates . . . . .                       | 9         |
| 4.2 Arbitrary powers of geometric products . . . . .                                | 10        |
| 4.3 A determinant and the recovery of logarithmic data . . . . .                    | 11        |
| <b>5 Partial fractions and divisor identities</b>                                   | <b>12</b> |
| 5.1 An interpolation formula for complete symmetric functions . . . . .             | 12        |
| 5.2 A finite divisor-sum master identity . . . . .                                  | 12        |
| 5.3 The ordinary harmonic limit . . . . .                                           | 13        |
| <b>6 Jackson differentiation and geometric Newton interpolation</b>                 | <b>14</b> |
| 6.1 The operator and its commutation rule . . . . .                                 | 14        |
| 6.2 A geometric Taylor basis . . . . .                                              | 14        |
| 6.3 Explicit stencils and divided differences . . . . .                             | 15        |
| <b>7 Two <math>q</math>-Stirling arrays and normal ordering</b>                     | <b>17</b> |
| 7.1 The monic $q$ -integer factorial basis . . . . .                                | 17        |
| 7.2 A Gaussian closed sum, with every power of $q$ accounted for . . . . .          | 18        |
| 7.3 Bell formulae for every diagonal . . . . .                                      | 18        |
| 7.4 The normal-ordering normalization . . . . .                                     | 19        |
| <b>8 Weighted Bell polynomials and Riordan composition</b>                          | <b>20</b> |
| 8.1 Gaussian convolution as a change of coefficient coordinates . . . . .           | 20        |
| 8.2 Composition coefficients without an ambiguous “ $q$ -Bell” convention . . . . . | 21        |
| 8.3 Weighted Riordan arrays . . . . .                                               | 21        |
| 8.4 Why Jackson composition is a different problem . . . . .                        | 22        |

|                                                                             |           |
|-----------------------------------------------------------------------------|-----------|
| <b>9 Lagrange inversion with geometric product data</b>                     | <b>23</b> |
| 9.1 The ordinary inversion theorem over a $q$ -dependent field . . . . .    | 23        |
| 9.2 A recurrence-free inverse for a product of shifted factorials . . . . . | 23        |
| 9.3 A two-variable inverse with its Jacobian factor exposed . . . . .       | 24        |
| <b>10 The coefficients of a Gaussian polynomial</b>                         | <b>26</b> |
| 10.1 A divisor-sum formula for each coefficient . . . . .                   | 26        |
| 10.2 A second finite formula by inclusion–exclusion . . . . .               | 27        |
| 10.3 Finite Fourier extraction . . . . .                                    | 27        |
| <b>11 Parameter derivatives and the expansion at one</b>                    | <b>28</b> |
| 11.1 Why the logarithmic parameter is the natural coordinate . . . . .      | 28        |
| 11.2 Exact coefficient moments and probability . . . . .                    | 29        |
| 11.3 Arbitrary parameter points and Eulerian rational functions . . . . .   | 30        |
| <b>12 Cyclotomic structure and all-order root-of-unity jets</b>             | <b>31</b> |
| 12.1 Cyclotomic factorization and carries . . . . .                         | 31        |
| 12.2 The finite $q$ -Lucas theorem . . . . .                                | 31        |
| 12.3 Deflating all apparent singularities . . . . .                         | 32        |
| <b>13 Weighted sums and differentiated identities</b>                       | <b>34</b> |
| 13.1 Polynomial weights in the summation index . . . . .                    | 34        |
| 13.2 Alternating sums at a product zero . . . . .                           | 34        |
| 13.3 An all-order differentiation of central Vandermonde . . . . .          | 35        |
| <b>14 Gaussian filters on geometric grids</b>                               | <b>37</b> |
| 14.1 Spectral design instead of recursive extrapolation . . . . .           | 37        |
| 14.2 Noise amplification is bounded at a fixed geometric ratio . . . . .    | 38        |
| 14.3 Fractional powers and arithmetic exponent lattices . . . . .           | 39        |
| 14.4 Logarithmic errors and repeated spectral roots . . . . .               | 39        |
| 14.5 Connection with the Fabius-function setting . . . . .                  | 40        |
| <b>15 Multinomial coefficients and factorial ratios</b>                     | <b>41</b> |
| 15.1 The Gaussian multinomial extension . . . . .                           | 41        |
| 15.2 A polynomiality criterion for balanced factorial ratios . . . . .      | 42        |
| <b>16 Computation, verification, and integration</b>                        | <b>44</b> |
| 16.1 Choosing a representation for the task . . . . .                       | 44        |
| 16.2 Arithmetic cost and finite verification . . . . .                      | 44        |
| 16.3 A theorem-level insertion map . . . . .                                | 45        |
| 16.4 Concluding perspective . . . . .                                       | 46        |
| <b>A A compact formula atlas</b>                                            | <b>47</b> |
| <b>B Source provenance and reproducibility</b>                              | <b>49</b> |
| B.1 Source and scope . . . . .                                              | 49        |
| B.2 Files and commands . . . . .                                            | 49        |
| <b>Bibliography</b>                                                         | <b>50</b> |

# Chapter 1

## The coefficient-calculus setting

### 1.1 Rings, indices, and specialization

Unless stated otherwise,  $q$  is an indeterminate and calculations with quotients take place over  $\mathbb{Q}(q)$ , extended by the finitely many auxiliary parameters that occur. A formal series in  $z$  is an element of  $K[[z]]$  for a commutative  $\mathbb{Q}$ -algebra  $K$ . Composition  $A(B(z))$  is defined when  $B(0) = 0$ ; inversion under multiplication requires an invertible constant term. These statements have no convergence content.

All integer indices are nonnegative unless explicitly qualified. Empty products are one, sums over empty ranges are zero, and  $0^0 = 1$  in coefficient formulae. Gaussian coefficients outside  $0 \leq k \leq n$  are zero. Define

$$(a; q)_N = \prod_{j=0}^{N-1} (1 - aq^j), \quad [n]_q = \sum_{j=0}^{n-1} q^j, \quad [n]_q! = \prod_{j=1}^n [j]_q. \quad (1.1)$$

In particular,  $[0]_q = 0$  and  $[0]_q! = 1$ . The polynomial interpretation of  $[n]_q$  makes  $[n]_1 = n$  immediate. The quotient  $(1 - q^n)/(1 - q)$  is a useful representation, not the definition at  $q = 1$ . Our  $q$ -shifted factorial convention agrees with [2].

When a rational representation equals a polynomial, specialization means specializing that polynomial, after cancellation. In particular, the expression  $(q; q)_n / ((q; q)_k (q; q)_{n-k})$  must not be evaluated as an uncanceled  $0/0$  at a root of unity. Later we give explicit polynomial and local-series methods that do not make this mistake.

### 1.2 Bell and Bernoulli normalizations

The complete and partial exponential Bell polynomials are characterized by

$$\exp \left( \sum_{r \geq 1} x_r \frac{z^r}{r!} \right) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}, \quad (1.2)$$

$$\frac{1}{k!} \left( \sum_{r \geq 1} x_r \frac{z^r}{r!} \right)^k = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{z^n}{n!}. \quad (1.3)$$

Thus  $\mathcal{B}_0 = 1$ , and

$$\mathcal{B}_n(x_1, \dots, x_n) = n! \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum r m_r = n}} \prod_{r=1}^n \frac{1}{m_r!} \left( \frac{x_r}{r!} \right)^{m_r}. \quad (1.4)$$

For  $\mathcal{B}_{n,k}$ , impose also  $\sum m_r = k$ . Expanding each exponential in (1.2) proves (1.4). In particular, a formula in Bell polynomials is a finite-sum formula, not a hidden recurrence.

Bernoulli numbers are denoted  $\beta_r$ , with

$$\frac{t}{e^t - 1} = \sum_{r \geq 0} \beta_r \frac{t^r}{r!}, \quad \beta_1 = -\frac{1}{2}. \quad (1.5)$$

We use  $s(n, k)$  for signed ordinary Stirling numbers of the first kind and  $S(n, k)$  for ordinary Stirling numbers of the second kind. Their  $q$ -versions will be defined independently, with the normalization stated in the definition.

### 1.3 Three distinct operations

There are three operations that should not be conflated. One may specialize the variables of a symmetric polynomial to a geometric progression. One may replace an ordinary derivative by the Jackson derivative. Or one may merely record ordinary series coefficients using  $[n]_q!$  rather than  $n!$ . All three lead to Gaussian coefficients, but their product and composition rules are not interchangeable.

#### Master principle

The robust route is

finite alphabet  $\longrightarrow$  power sums  $\longrightarrow$  formal logarithm  $\longrightarrow$  Bell coefficients.

Operator-adapted interpolation and formal reversion can then be attached to this route without guessing which ordinary factorials should acquire a subscript  $q$ .

## Chapter 2

# Gaussian polynomials and finite products

### 2.1 A division-free definition

**Definition 2.1.** For  $0 \leq k \leq n$ , set

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{0 \leq i_1 < \dots < i_k < n} q^{i_1 + \dots + i_k - \binom{k}{2}}. \quad (2.1)$$

The  $k = 0$  summand is the empty subset and contributes one.

Every exponent is nonnegative, because  $i_j \geq j - 1$ . Consequently this is a polynomial in  $\mathbb{Z}_{\geq 0}[q]$ , valid at every complex  $q$ .

**Theorem 2.2** (Pascal, factorial, and symmetry formulae). *For  $n \geq 1$ , the Gaussian polynomials satisfy the first two recurrences below; the remaining identities hold for  $0 \leq k \leq n$ :*

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + q^{n-k} \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q, \quad (2.2)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = q^k \begin{bmatrix} n-1 \\ k \end{bmatrix}_q + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}_q, \quad (2.3)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}} = \frac{[n]_q!}{[k]_q! [n-k]_q!}, \quad (2.4)$$

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \begin{bmatrix} n \\ n-k \end{bmatrix}_q, \quad \begin{bmatrix} n \\ k \end{bmatrix}_{q^{-1}} = q^{-k(n-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q. \quad (2.5)$$

*The quotient identities hold in  $\mathbb{Q}(q)$ . For  $0 \leq k \leq n$ , the polynomial has degree  $k(n-k)$ , constant coefficient one, and leading coefficient one.*

*Proof.* Separate the subsets in (2.1) according as they contain  $n-1$ . Removing that element decreases the exponent by  $n-k$ , proving (2.2). The factorial quotient has the same boundary values and satisfies the same recurrence: after putting the two terms over a common denominator the remaining numerator is  $(1 - q^{n-k}) + q^{n-k}(1 - q^k) = 1 - q^n$ . Induction therefore proves (2.4). Its symmetry gives (2.3) by replacing  $k$  with  $n-k$ .

The smallest subset  $\{0, \dots, k-1\}$  has exponent zero, and the largest  $\{n-k, \dots, n-1\}$  has exponent  $k(n-k)$ . Each is unique. Reflecting indices by  $i_j \mapsto n-1-i_{k+1-j}$  replaces an exponent  $d$  by  $k(n-k) - d$ , proving reciprocity and palindromicity.  $\square$



A subset gives a partition inside a  $k$ -by- $(n - k)$  rectangle by taking  $\lambda_j = i_{k+1-j} - (k - j)$ . These integers decrease weakly, lie between zero and  $n - k$ , and sum to the exponent in (2.1). The construction is reversible. This proves the rectangular-partition interpretation directly. At  $q = 1$ , every subset has weight one, so  $\left[ \begin{smallmatrix} n \\ k \end{smallmatrix} \right]_1 = \binom{n}{k}$ .

**Example 2.3.**

$$\left[ \begin{smallmatrix} 4 \\ 2 \end{smallmatrix} \right]_q = 1 + q + 2q^2 + q^3 + q^4, \quad \left[ \begin{smallmatrix} 6 \\ 3 \end{smallmatrix} \right]_q = 1 + q + 2q^2 + 3q^3 + 3q^4 + 3q^5 + 3q^6 + 2q^7 + q^8 + q^9.$$

The coefficient sums are respectively 6 and 20; the symmetry centers are respectively 2 and  $9/2$ .

## 2.2 The finite and reciprocal binomial theorems

**Theorem 2.4.** For every  $N \geq 0$ ,

$$(z; q)_N = \sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \left[ \begin{smallmatrix} N \\ k \end{smallmatrix} \right]_q z^k. \quad (2.6)$$

For  $N \geq 1$ , as a formal series in  $z$ ,

$$\frac{1}{(z; q)_N} = \sum_{r \geq 0} \left[ \begin{smallmatrix} N + r - 1 \\ r \end{smallmatrix} \right]_q z^r. \quad (2.7)$$

When  $N = 0$ , the reciprocal is one.

*Proof.* To obtain  $z^k$  in the finite product, choose the  $-zq^i$  term in precisely  $k$  factors. Its weight is  $(-1)^k q^{\sum i}$ ; comparison with (2.1) proves (2.6).

Write  $H_N(z) = 1/(z; q)_N$ . Then  $(1 - q^{N-1}z)H_N(z) = H_{N-1}(z)$ . If  $c_{N,r} = [z^r]H_N$ , it follows that  $c_{N,r} = c_{N-1,r} + q^{N-1}c_{N,r-1}$ . The candidate  $\left[ \begin{smallmatrix} N + r - 1 \\ r \end{smallmatrix} \right]_q$  has this recurrence by (2.2), with  $c_{1,r} = 1$  and  $c_{N,0} = 1$ . Induction on  $N + r$  proves the reciprocal identity.  $\square$

These classical identities are the finite and negative-power backbone of the extension; compare the standard conventions in [2, 3].

## 2.3 A formal infinite-binomial identity without a topology error

**Theorem 2.5.** There is a unique  $R_a(z) \in \mathbb{Q}(q, a)[[z]]$  with  $R_a(0) = 1$  satisfying

$$(1 - z)R_a(z) = (1 - az)R_a(qz). \quad (2.8)$$

It is

$$R_a(z) = \sum_{r \geq 0} \frac{(a; q)_r}{(q; q)_r} z^r. \quad (2.9)$$

For complex  $|q| < 1$  and  $|z| < 1$ , it also equals

$$R_a(z) = \frac{(az; q)_\infty}{(z; q)_\infty}. \quad (2.10)$$

*Proof.* Comparison of coefficients in (2.8) gives  $(1 - q^r)c_r = (1 - aq^{r-1})c_{r-1}$ , with  $c_0 = 1$ . This proves both uniqueness and (2.9). For  $|q| < 1$ , the infinite products converge locally uniformly in their arguments, since the tails are bounded by convergent geometric sums. The denominator has no zero for  $|z| < 1$ . Their ratio satisfies the same functional equation, is analytic at zero, and has constant term one. Its Taylor coefficients are therefore the coefficients already computed. Unless the series terminates, the coefficient ratio tends to one, so the series converges absolutely for  $|z| < 1$ ; termination makes convergence immediate.  $\square$

#### Normalization and domain

With  $q$  merely an indeterminate, the product  $\prod_{j \geq 0} (1 - zq^j)$  is not defined by  $z$ -adic convergence: its linear coefficient changes at every step. Equation (2.9) or the functional equation is the formal definition. Equation (2.10) is an additional analytic realization.

## Chapter 3

# Convolution, inversion, and linearization

### 3.1 Two-block and many-block Vandermonde identities

**Theorem 3.1.** For  $m, n, r \geq 0$ ,

$$\begin{bmatrix} n+m \\ r \end{bmatrix}_q = \sum_{k=\max(0, r-m)}^{\min(n, r)} q^{(n-k)(r-k)} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} m \\ r-k \end{bmatrix}_q. \quad (3.1)$$

More generally, if  $n_1 + \dots + n_s = N$ , then

$$\begin{bmatrix} N \\ r \end{bmatrix}_q = \sum_{\substack{0 \leq k_i \leq n_i \\ \sum k_i = r}} q^{\sum_{i < j} (n_i - k_i) k_j} \prod_{i=1}^s \begin{bmatrix} n_i \\ k_i \end{bmatrix}_q. \quad (3.2)$$

*Proof.* Split  $(-z; q)_{n+m} = (-z; q)_n (-zq^n; q)_m$  and use (2.6) with  $-z$  in place of  $z$ . After dividing the coefficient of  $z^r$  by  $q^{\binom{r}{2}}$ , the exponent is

$$\binom{k}{2} + \binom{r-k}{2} + n(r-k) - \binom{r}{2} = (n-k)(r-k).$$

The same computation for consecutive blocks of lengths  $n_i$  gives  $\sum_i \binom{k_i}{2} + \sum_{i < j} n_i k_j - \binom{r}{2} = \sum_{i < j} (n_i - k_i) k_j$ , proving the general case.  $\square$

In particular, symmetry and the substitution  $k \mapsto n - k$  give

$$\sum_{k=0}^n q^{k^2} \begin{bmatrix} n \\ k \end{bmatrix}_q^2 = \begin{bmatrix} 2n \\ n \end{bmatrix}_q. \quad (3.3)$$

The exponent  $k^2$  is essential; removing it does not give a Gaussian analogue of the ordinary central-binomial identity.

### 3.2 Gaussian inversion and the subspace lattice

**Theorem 3.2** (Gaussian binomial inversion). For arbitrary sequences over  $\mathbb{Q}(q)$ ,

$$b_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k \iff a_n = \sum_{k=0}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q b_k. \quad (3.4)$$

Both identities also make sense over  $\mathbb{Z}[q]$  when the sequences do.

*Proof.* The coefficient of  $a_j$  after substitution on the right is

$$\sum_{k=j}^n (-1)^{n-k} q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q.$$

Cancel factorials to obtain  $\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} k \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_q \begin{bmatrix} n-j \\ k-j \end{bmatrix}_q$ . The remaining sum is  $(1; q)_{n-j}$ , after reversing its index and using symmetry. It is zero unless  $n = j$ , when it is one. The two triangular matrices therefore multiply to the identity. Their unit diagonal makes either one-sided inverse also the inverse on every finite truncation.  $\square$

The same inverse has a geometric interpretation. For a prime power  $Q$ , the number of  $k$ -dimensional subspaces of  $\mathbb{F}_Q^n$  is  $\begin{bmatrix} n \\ k \end{bmatrix}_Q$ . Indeed, count ordered linearly independent  $k$ -tuples and divide by the number of ordered bases of  $\mathbb{F}_Q^k$ :

$$\frac{\prod_{j=0}^{k-1} (Q^n - Q^j)}{\prod_{j=0}^{k-1} (Q^k - Q^j)} = \begin{bmatrix} n \\ k \end{bmatrix}_Q.$$

An interval of rank difference  $r$  in the subspace lattice is itself a subspace lattice of dimension  $r$ . Its incidence-algebra Möbius function is  $(-1)^r Q^{\binom{r}{2}}$ , because its defining convolution with the constant-one function is exactly  $(1; Q)_r = 0$  for  $r > 0$ . This gives a direct counterpart of the source article's use of incidence algebras for ordinary inversion.

### 3.3 A positive linearization formula

**Theorem 3.3.** For  $0 \leq k, \ell \leq n$ ,

$$\begin{bmatrix} n \\ k \end{bmatrix}_q \begin{bmatrix} n \\ \ell \end{bmatrix}_q = \sum_{r=\max(k, \ell)}^{\min(n, k+\ell)} q^{(r-k)(r-\ell)} \begin{bmatrix} n \\ r \end{bmatrix}_q \begin{bmatrix} r \\ k \end{bmatrix}_q \begin{bmatrix} k \\ k+\ell-r \end{bmatrix}_q. \quad (3.5)$$

*Proof.* First let  $q = Q$  be a prime power and count ordered pairs  $(V, W)$  of subspaces of dimensions  $k, \ell$ . Set  $U = V + W$ ,  $\dim U = r$ , and  $J = V \cap W$ ,  $\dim J = j = k + \ell - r$ . Choose  $U$ , then  $V \subseteq U$ , then  $J \subseteq V$ , giving the three Gaussian factors. In  $U/J$ , the space  $V/J$  has dimension  $k - j = r - \ell$ . The possible  $W/J$  are complements of this space, of dimension  $\ell - j = r - k$ . Relative to one fixed complement, every other complement is the graph of a unique linear map into  $V/J$ . There are  $Q^{(r-k)(r-\ell)}$  such maps. This proves the formula for every prime power. Both sides are polynomials in  $q$ ; infinitely many equal values prove the polynomial identity.  $\square$

For example,

$$[n]_q^2 = [n]_q + q(1+q) \begin{bmatrix} n \\ 2 \end{bmatrix}_q. \quad (3.6)$$

This is a Gaussian factorial-moment identity. A general power analogue follows from  $q$ -Stirling basis change in Chapter 7.

### 3.4 Pochhammer convolution and terminating hypergeometric sums

**Theorem 3.4.** For  $N \geq 0$ ,

$$(ab; q)_N = \sum_{j=0}^N \begin{bmatrix} N \\ j \end{bmatrix}_q a^j (b; q)_j (a; q)_{N-j}. \quad (3.7)$$

Consequently, wherever the displayed denominators are nonzero,

$$\sum_{j=0}^N \frac{(q^{-N}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} q^j = a^N \frac{(c/a; q)_N}{(c; q)_N}, \quad (3.8)$$

$$\sum_{j=0}^N \frac{(q^{-N}; q)_j (a; q)_j}{(q; q)_j (c; q)_j} \left( \frac{cq^N}{a} \right)^j = \frac{(c/a; q)_N}{(c; q)_N}. \quad (3.9)$$

The last two formulae are the terminating  $q$ -Chu–Vandermonde identities.

*Proof.* The functional equation (2.8) implies  $R_{ab}(z) = R_a(z)R_b(az)$ : both sides have constant term one and satisfy  $(1-z)R(z) = (1-abz)R(qz)$ . Comparison of  $z^N$  in (2.9), followed by multiplication by  $(q; q)_N$ , proves (3.7). This proof is entirely formal.

Divide that formula by  $(a; q)_N$ . Direct reversal of finite products gives

$$\begin{bmatrix} N \\ j \end{bmatrix}_q \frac{(a; q)_{N-j}}{(a; q)_N} a^j = \frac{(q^{-N}; q)_j}{(q; q)_j (q^{1-N}/a; q)_j} q^j. \quad (3.10)$$

For completeness, use

$$(q^{-N}; q)_j = (-1)^j q^{-Nj + \binom{j}{2}} \frac{(q; q)_N}{(q; q)_{N-j}}$$

and

$$(aq^{N-j}; q)_j = (-a)^j q^{Nj - j(j+1)/2} (q^{1-N}/a; q)_j$$

in the left side. The powers reduce to  $q^j$ , proving (3.10). Now set the original  $a = q^{1-N}/c$  and rename the original  $b$  as  $a$ . The product reversal  $(x; q)_N = (-x)^N q^{\binom{N}{2}} (q^{1-N}/x; q)_N$  turns  $(ab; q)_N/(a; q)_N$  into the right side of (3.8).

Finally replace  $q, a, c$  in (3.8) by  $q^{-1}, a^{-1}, c^{-1}$ . Reversing each finite product changes the summand's extra power into  $(cq^N/a)^j$ ; the prefactor  $a^{-N}$  on the right cancels the reversal factor  $a^N$ . This yields (3.9). All calculations first take place in a rational-function field; any admissible specialization then follows.  $\square$

The identities are classical; [4] records their hypergeometric normalizations. The proof above explains why they are coefficient-convolution identities before they are hypergeometric summations.

## Chapter 4

# Symmetric functions and the Bell master formula

### 4.1 Elementary, complete, and power-sum coordinates

For a finite alphabet  $X = (x_1, \dots, x_N)$ , define

$$e_r(X) = \sum_{i_1 < \dots < i_r} x_{i_1} \cdots x_{i_r}, \quad h_r(X) = \sum_{i_1 \leq \dots \leq i_r} x_{i_1} \cdots x_{i_r}, \quad p_r(X) = \sum_i x_i^r.$$

Both degree-zero functions are one. Direct multiplication gives

$$\sum_{r \geq 0} e_r(X) z^r = \prod_i (1 + x_i z), \quad (4.1)$$

$$\sum_{r \geq 0} h_r(X) z^r = \prod_i (1 - x_i z)^{-1}. \quad (4.2)$$

Taking formal logarithms gives, for  $r \geq 0$ ,

$$e_r(X) = \frac{1}{r!} \mathcal{B}_r(p_1, -1!p_2, 2!p_3, \dots, (-1)^{r-1}(r-1)!p_r), \quad (4.3)$$

$$h_r(X) = \frac{1}{r!} \mathcal{B}_r(p_1, 1!p_2, 2!p_3, \dots, (r-1)!p_r). \quad (4.4)$$

These identities follow from (1.2); no additional combinatorial normalization is involved. Also  $\sum_{j=0}^r (-1)^j e_j(X) h_{r-j}(X) = \delta_{r0}$ , because the two generating functions with opposite signs multiply to one.

**Theorem 4.1** (Principal specialization). *For  $X_N = (1, q, \dots, q^{N-1})$ ,*

$$p_r(X_N) = [N]_{q^r}, \quad e_k(X_N) = q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q, \quad h_k(X_N) = \begin{bmatrix} N+k-1 \\ k \end{bmatrix}_q \quad (N \geq 1). \quad (4.5)$$

*Proof.* The first formula is a geometric sum. The other two follow by comparing (4.1) and (4.2) with (2.6) and (2.7).  $\square$

Thus a Gaussian coefficient is an ordinary symmetric polynomial at geometric nodes. In particular,

$$q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q = \frac{1}{k!} \mathcal{B}_k \left( ((-1)^{r-1}(r-1)![N]_{q^r})_{r=1}^k \right), \quad (4.6)$$

$$\begin{bmatrix} N+k-1 \\ k \end{bmatrix}_q = \frac{1}{k!} \mathcal{B}_k \left( ((r-1)![N]_{q^r})_{r=1}^k \right). \quad (4.7)$$

For instance,

$$\begin{bmatrix} N+1 \\ 2 \end{bmatrix}_q = \frac{[N]_q^2 + [N]_q^2}{2}, \quad q \begin{bmatrix} N \\ 2 \end{bmatrix}_q = \frac{[N]_q^2 - [N]_q^2}{2}.$$

The apparent halves cancel because these are symmetric-function identities over  $\mathbb{Z}[q]$ .

## 4.2 Arbitrary powers of geometric products

**Theorem 4.2** (Geometric-product coefficient formula). *Let  $N_\ell \geq 0$ ,  $d_\ell \geq 1$ , and let  $a_\ell, \alpha_\ell$  be commuting parameters in a  $\mathbb{Q}$ -algebra. Define formally*

$$F(z) = \prod_{\ell=1}^L (a_\ell z; q^{d_\ell})_{N_\ell}^{-\alpha_\ell}, \quad P_r = \sum_{\ell=1}^L \alpha_\ell a_\ell^r [N_\ell]_{q^{d_\ell}}. \quad (4.8)$$

*Powers with nonintegral exponents mean  $\exp(-\alpha_\ell \log(\cdot))$  with constant term one. Then*

$$[z^n]F(z) = \frac{1}{n!} \mathcal{B}_n(P_1, 1!P_2, \dots, (n-1)!P_n) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum r m_r = n}} \prod_{r=1}^n \frac{P_r^{m_r}}{r^{m_r} m_r!}. \quad (4.9)$$

*Proof.* Each factor has a formal logarithm, and

$$\log F(z) = \sum_{\ell=1}^L \alpha_\ell \sum_{j=0}^{N_\ell-1} \sum_{r \geq 1} \frac{(a_\ell q^{d_\ell j} z)^r}{r} = \sum_{r \geq 1} P_r \frac{z^r}{r}.$$

At every fixed degree this rearranges only finitely many contributions from the finite alphabets. Apply (1.2) with  $x_r = (r-1)!P_r$ , then (1.4).  $\square$

The first four coefficients are

$$\begin{aligned} f_1 &= P_1, & f_2 &= \frac{P_1^2 + P_2}{2}, \\ f_3 &= \frac{P_1^3 + 3P_1P_2 + 2P_3}{6}, \\ f_4 &= \frac{P_1^4 + 6P_1^2P_2 + 3P_2^2 + 8P_1P_3 + 6P_4}{24}. \end{aligned}$$

Negative integral  $\alpha_\ell$  encode numerator products, positive integral ones encode repeated reciprocal products, and arbitrary parameters encode binomial powers. Several bases  $q^{d_\ell}$  require no new machinery.

**Corollary 4.3** (A two-base convolution). *For  $M, N \geq 1$ ,*

$$\begin{aligned} \sum_{j=0}^n a^j b^{n-j} \begin{bmatrix} M+j-1 \\ j \end{bmatrix}_q \begin{bmatrix} N+n-j-1 \\ n-j \end{bmatrix}_{q^d} \\ = \frac{1}{n!} \mathcal{B}_n \left( ((r-1)!(a^r[M]_{q^r} + b^r[N]_{q^{dr}}))_{r=1}^n \right). \end{aligned} \quad (4.10)$$

*Proof.* Expand the two reciprocal products using (2.7) and multiply. Alternatively, apply Theorem 4.2 directly to their product. Equating the two coefficients proves the identity.  $\square$

### 4.3 A determinant and the recovery of logarithmic data

**Proposition 4.4.** *If  $F(z) = \exp(\sum_{r \geq 1} P_r z^r / r) = \sum f_n z^n$ , then*

$$f_n = \frac{1}{n!} \det \begin{pmatrix} P_1 & -1 & 0 & \cdots & 0 \\ P_2 & P_1 & -2 & \cdots & 0 \\ P_3 & P_2 & P_1 & \ddots & \vdots \\ \vdots & \vdots & \ddots & \ddots & -(n-1) \\ P_n & P_{n-1} & \cdots & P_2 & P_1 \end{pmatrix}. \quad (4.11)$$

*Conversely, put  $x_j = j!f_j$ . Then*

$$P_n = \frac{1}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}). \quad (4.12)$$

*Proof.* The equation  $F' = (\sum_{r \geq 1} P_r z^{r-1})F$  gives  $nf_n = \sum_{r=1}^n P_r f_{n-r}$ . If  $D_n$  is the determinant above, expansion along its last row gives

$$D_n = \sum_{r=1}^n \frac{(n-1)!}{(n-r)!} P_r D_{n-r}, \quad D_0 = 1.$$

The superdiagonal products supply the factorial ratio, and the Laplace signs cancel their signs. Therefore  $D_n/n!$  satisfies the same recurrence and initial value as  $f_n$ . For the inverse formula, expand  $\log F = \sum_{k \geq 1} (-1)^{k-1} (F-1)^k / k$ , use (1.3), and compare  $z^n$  with  $P_n/n$ .  $\square$

This is precisely the moment-cumulant mechanism in the source article, with geometric power sums serving as the logarithmic data.



## Chapter 5

# Partial fractions and divisor identities

### 5.1 An interpolation formula for complete symmetric functions

**Lemma 5.1.** *If  $x_1, \dots, x_N$  are distinct and nonzero, then*

$$\prod_{i=1}^N \frac{1}{1 - x_i z} = \sum_{i=1}^N \frac{x_i^{N-1}}{\prod_{j \neq i} (x_i - x_j)} \frac{1}{1 - x_i z}. \quad (5.1)$$

Consequently,

$$h_r(x_1, \dots, x_N) = \sum_{i=1}^N \frac{x_i^{r+N-1}}{\prod_{j \neq i} (x_i - x_j)}. \quad (5.2)$$

*Proof.* Multiply the proposed rational identity by  $\prod_j (1 - x_j z)$ . Both sides are polynomials of degree at most  $N - 1$ ; at  $z = 1/x_i$ , only one summand on the right survives, and it equals one. Agreement at  $N$  distinct points proves the identity. Expanding each geometric denominator and comparing coefficients proves (5.2).  $\square$

The divided-difference expression on the right has removable singularities when nodes coincide; the left side supplies the polynomial continuation. This is a simple instance of why a rational summation formula can encode a polynomial.

### 5.2 A finite divisor-sum master identity

**Theorem 5.2** (Dilcher-type identity and Bell form). *For  $N \geq 1$ ,  $r \geq 0$ , and indeterminate  $q$ , let  $u_j = q^j / (1 - q^j)$ . Then*

$$\sum_{j=1}^N (-1)^{j-1} \frac{q^{\binom{j}{2} + rj}}{(1 - q^j)^r} \left[ \begin{matrix} N \\ j \end{matrix} \right]_q = h_r(u_1, \dots, u_N) \quad (5.3)$$

$$\begin{aligned} &= \sum_{1 \leq i_1 \leq \dots \leq i_r \leq N} \frac{q^{i_1 + \dots + i_r}}{(1 - q^{i_1}) \dots (1 - q^{i_r})} \\ &= \frac{1}{r!} \mathcal{B}_r \left( \left( (s-1)! \sum_{j=1}^N \frac{q^{js}}{(1 - q^j)^s} \right)_{s=1}^r \right). \end{aligned} \quad (5.4)$$

For  $r = 0$ , every right-hand side is interpreted as one.

*Proof.* Apply Theorem 5.1 to  $q, q^2, \dots, q^N$ . For the node  $q^j$ , the residue coefficient is

$$\frac{1}{\prod_{i \neq j} (1 - q^{i-j})} = \frac{(-1)^{j-1} q^{\binom{j}{2}}}{(q; q)_{j-1} (q; q)_{N-j}}.$$

Multiplication by  $(q; q)_N$  therefore yields

$$\frac{(q; q)_N}{(zq; q)_N} = \sum_{j=1}^N (-1)^{j-1} q^{\binom{j}{2}} (1 - q^j) \begin{bmatrix} N \\ j \end{bmatrix}_q \frac{1}{1 - zq^j}. \quad (5.5)$$

Set  $z = 1 + t$ . The left side becomes  $\prod_{j=1}^N (1 - u_j t)^{-1}$ . The coefficient of  $t^r$  on the right is exactly the left side of (5.3). The complete-symmetric and Bell descriptions now follow from (4.2) and (4.4).  $\square$

This is the finite identity associated with Dilcher's divisor-function identities [6]. The partial-fraction proof simultaneously explains the alternating single sum, the weakly ordered multiple sum, and the Bell polynomial. For  $r = 1$ , it reduces to

$$\sum_{j=1}^N (-1)^{j-1} \frac{q^{j(j+1)/2}}{1 - q^j} \begin{bmatrix} N \\ j \end{bmatrix}_q = \sum_{j=1}^N \frac{q^j}{1 - q^j}.$$

For  $r = 2$ , the right side is  $\frac{1}{2}((\sum u_j)^2 + \sum u_j^2)$ , and for  $r = 3$ , it is  $\frac{1}{6}((\sum u_j)^3 + 3(\sum u_j)(\sum u_j^2) + 2 \sum u_j^3)$ .

### 5.3 The ordinary harmonic limit

Multiply (5.3) by  $(1 - q)^r$  and let  $q \rightarrow 1$ . Since  $(1 - q)u_j \rightarrow 1/j$ , the limit is

$$\sum_{j=1}^N (-1)^{j-1} \binom{N}{j} \frac{1}{j^r} = h_r \left( 1, \frac{1}{2}, \dots, \frac{1}{N} \right) = \frac{1}{r!} \mathcal{B}_r(H_N^{(1)}, 1!H_N^{(2)}, \dots, (r-1)!H_N^{(r)}), \quad (5.6)$$

where  $H_N^{(s)} = \sum_{j=1}^N j^{-s}$ . This limit is termwise because all sums are finite. Thus the ordinary binomial-harmonic identities of coefficient calculus are literal degenerations of the Gaussian identities, not merely analogous-looking formulae.

## Chapter 6

# Jackson differentiation and geometric Newton interpolation

### 6.1 The operator and its commutation rule

Let  $T_q f(x) = f(qx)$ , and let  $X$  denote multiplication by  $x$ . Define

$$D_q f(x) = \frac{f(x) - f(qx)}{(1-q)x}. \quad (6.1)$$

For polynomials, this has no singularity at  $x = 0$ , because the numerator has zero constant term. It acts by  $D_q x^m = [m]_q x^{m-1}$ , and hence

$$D_q X = 1 + qX D_q, \quad D_q T_q = qT_q D_q. \quad (6.2)$$

These identities follow on every monomial and hence on every polynomial. This is the convention for the Jackson derivative used throughout; compare [2].

**Theorem 6.1** (The shifted product rule). *For  $n \geq 0$ ,*

$$D_q^n (fg)(x) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (D_q^k f)(q^{n-k}x)(D_q^{n-k} g)(x). \quad (6.3)$$

*Proof.* The first-order formula is  $D_q(fg) = (D_q f)g + (T_q f)D_q g$ , obtained by adding and subtracting  $f(qx)g(x)$  in the numerator. Suppose the formula is true at order  $n$ . Differentiating a summand gives a contribution with  $k$  unchanged and a contribution with  $k$  increased by one. In the latter, differentiation of  $(D_q^k f)(q^{n-k}x)$  produces a factor  $q^{n-k}$ . Thus the new coefficient at index  $k$  is

$$\begin{bmatrix} n \\ k \end{bmatrix}_q + q^{n-k+1} \begin{bmatrix} n \\ k-1 \end{bmatrix}_q = \begin{bmatrix} n+1 \\ k \end{bmatrix}_q, \text{ by (2.2). The shifts are exactly those in the stated formula. } \quad \square$$

#### Normalization and domain

The argument shift  $q^{n-k}x$  is part of the theorem. Replacing the ordinary binomial coefficient by a Gaussian coefficient in the usual Leibniz rule, while leaving all factors evaluated at  $x$ , is false.

### 6.2 A geometric Taylor basis

Define

$$(x-a)_q^{[k]} = \prod_{j=0}^{k-1} (x - aq^j). \quad (6.4)$$

This is a polynomial basis in  $x$ , not a power of  $x - a$ .

**Theorem 6.2** (Jackson–Newton expansion). *For every polynomial  $p$  of degree at most  $d$ ,*

$$p(x) = \sum_{k=0}^d \frac{(D_q^k p)(a)}{[k]_q!} (x - a)_q^{[k]}. \quad (6.5)$$

*The identity first holds over  $\mathbb{Q}(q, a)$ , and specializes wherever its coefficients are defined. At  $q = 1$  it becomes ordinary Taylor expansion at  $a$ .*

*Proof.* The polynomial  $D_q(x - a)_q^{[k]}$  has degree  $k - 1$  and leading coefficient  $[k]_q$ . At each  $x = aq^j$ ,  $0 \leq j \leq k - 2$ , its two numerator terms vanish. Therefore

$$D_q(x - a)_q^{[k]} = [k]_q (x - a)_q^{[k-1]}.$$

This argument at generic  $a, q$  proves a polynomial identity and hence also covers  $a = 0$ . Expand  $p$  in this monic basis, apply  $D_q^k$ , and set  $x = a$ . Terms with index less than  $k$  vanish under differentiation; terms with larger index retain a factor  $x - a$ ; the index  $k$  term yields  $[k]_q!$  times its coefficient. This proves the formula.  $\square$

For example,

$$x^2 = a^2 + (1 + q)a(x - a) + (x - a)(x - aq),$$

and

$$x^3 = a^3 + [3]_q a^2(x - a) + [3]_q a(x - a)(x - aq) + (x - a)(x - aq)(x - aq^2).$$

At a sample node  $x = aq^n$ , factors with  $k > n$  vanish, and direct factorization gives the discrete Gaussian transform

$$p(aq^n) = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (a(q - 1))^k q^{\binom{k}{2}} (D_q^k p)(a). \quad (6.6)$$

Indeed,  $\prod_{j < k} (aq^n - aq^j) = [a(q - 1)]^k q^{\binom{k}{2}} [k]_q! \begin{bmatrix} n \\ k \end{bmatrix}_q$ . This identifies Gaussian inversion with the extraction of geometric Newton coefficients.

### 6.3 Explicit stencils and divided differences

**Theorem 6.3.** *For  $n \geq 0$ ,*

$$D_q^n f(x) = \frac{1}{(1 - q)^n x^n} \sum_{j=0}^n (-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q f(q^j x). \quad (6.7)$$

*At distinct nonzero nodes  $a, aq, \dots, aq^n$ ,*

$$f[a, aq, \dots, aq^n] = \frac{(D_q^n f)(a)}{[n]_q!}. \quad (6.8)$$

*The left side denotes the leading coefficient of the degree-at-most- $n$  interpolating polynomial through the given data.*

*Proof.* The commutation relation  $T_q X^{-1} = q^{-1} X^{-1} T_q$  gives

$$D_q^n = (1 - q)^{-n} X^{-n} \prod_{r=0}^{n-1} (1 - q^{-r} T_q).$$

Expand the finite product with base  $q^{-1}$ , and use reciprocity (2.5). Its  $T_q^j$  coefficient becomes  $(-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q$ , proving (6.7). Let  $p$  interpolate  $f$  at these nodes. Formula (6.7) shows that  $(D_q^n f)(a) = (D_q^n p)(a)$ . The latter equals  $[n]_q!$  times the leading coefficient of  $p$ , either by monomial differentiation or by Theorem 6.2. This proves (6.8).  $\square$

**Corollary 6.4** (An exact Gaussian coefficient extractor). *If  $p$  has degree at most  $n$ , then*

$$[x^n]p(x) = \frac{1}{(q; q)_n} \sum_{j=0}^n (-1)^j q^{\binom{j}{2} - (n-1)j} \begin{bmatrix} n \\ j \end{bmatrix}_q p(q^j). \quad (6.9)$$

*The numerator annihilates  $x^r$  for every  $0 \leq r < n$ .*

*Proof.* Apply (6.7) at  $x = 1$  and divide by  $[n]_q!$ . Alternatively, the numerator on  $x^r$  is  $(q^{r-n+1}; q)_n$ , by (2.6). One factor is zero for  $0 \leq r < n$ , while for  $r = n$  the product is  $(q; q)_n$ .  $\square$

This is the geometric counterpart of the additive finite-difference stencil in the source. Its cancelling objects are monomials evaluated at geometric nodes.

## Chapter 7

# Two $q$ -Stirling arrays and normal ordering

### 7.1 The monic $q$ -integer factorial basis

Define

$$P_k(x) = \prod_{j=0}^{k-1} (x - [j]_q), \quad P_n(x) = \sum_{k=0}^n s_q(n, k) x^k, \quad x^n = \sum_{k=0}^n S_q(n, k) P_k(x). \quad (7.1)$$

These are changes between two monic polynomial bases. They define  $s_q, S_q$  unambiguously over  $\mathbb{Z}[q]$ , even when specialized nodes coincide.

**Theorem 7.1.** *The arrays are mutually inverse, and*

$$s_q(n+1, k) = s_q(n, k-1) - [n]_q s_q(n, k), \quad (7.2)$$

$$S_q(n+1, k) = S_q(n, k-1) + [k]_q S_q(n, k), \quad (7.3)$$

$$s_q(n, k) = (-1)^{n-k} e_{n-k}([0]_q, \dots, [n-1]_q), \quad (7.4)$$

$$S_q(n, k) = h_{n-k}([0]_q, \dots, [k]_q). \quad (7.5)$$

Both arrays have entry one at  $(0, 0)$  and vanish outside the triangular range.

*Proof.* Multiplication of  $P_n(x)$  by  $x - [n]_q$  gives (7.2). The identity  $xP_k = P_{k+1} + [k]_q P_k$  gives (7.3). The coefficient formula (7.4) follows by choosing roots from the product defining  $P_n$ . For (7.5), use

$$h_r([0]_q, \dots, [k]_q) = h_r([0]_q, \dots, [k-1]_q) + [k]_q h_{r-1}([0]_q, \dots, [k]_q).$$

With  $r = n - k$ , this is precisely the recurrence for  $S_q$ , with the same boundary values. Finally, substituting either basis change into the other and using uniqueness of coefficients proves matrix inversion.  $\square$

At  $q = 1$ , these are the ordinary signed first-kind and second-kind Stirling numbers. At  $q = 0$ , all positive-index nodes equal one, so for  $n \geq 1$ ,

$$S_0(n, k) = \binom{n-1}{k-1}, \quad s_0(n, k) = (-1)^{n-k} \binom{n-1}{k-1} \quad (1 \leq k \leq n).$$

This follows from  $P_n(x) = x(x-1)^{n-1}$  and from the complete-symmetric formula. It illustrates the usefulness of a polynomial definition at degenerate bases.

## 7.2 A Gaussian closed sum, with every power of $q$ accounted for

**Theorem 7.2.** Over  $\mathbb{Q}(q)$ , for  $0 \leq k \leq n$ ,

$$S_q(n, k) = \frac{q^{-\binom{k}{2}}}{[k]_q!} \sum_{j=0}^k (-1)^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q [j]_q^n. \quad (7.6)$$

The expression has the polynomial continuation specified by (7.1).

*Proof.* The coefficient of the degree- $k$  Newton basis polynomial in the interpolation of  $x^n$  at  $[0]_q, \dots, [k]_q$  is

$$\sum_{j=0}^k \frac{[j]_q^n}{\prod_{0 \leq i \leq k, i \neq j} ([j]_q - [i]_q)}.$$

It equals the coefficient in the full Newton expansion because terms of degree larger than  $k$  vanish at all these nodes. For  $i < j$ ,  $[j]_q - [i]_q = q^i[j - i]_q$ , whereas for  $i > j$  it is  $-q^j[i - j]_q$ . Hence the denominator is

$$(-1)^{k-j} q^{jk-j(j+1)/2} [j]_q! [k-j]_q!.$$

Its inverse, multiplied by  $[k]_q!$ , has exponent  $\binom{j+1}{2} - jk = \binom{k-j}{2} - \binom{k}{2}$ . This proves (7.6).  $\square$

There is no guesswork in the factor  $q^{-\binom{k}{2}}$ : it is the normalization of the monic  $q$ -integer Newton basis. Omitting it produces a different array.

## 7.3 Bell formulae for every diagonal

For  $k \geq 0$  and  $r \geq 1$ , let

$$U_r(k) = \sum_{j=0}^k [j]_q^r = \frac{1}{(1-q)^r} \sum_{a=0}^r (-1)^a \binom{r}{a} [k+1]_q^a. \quad (7.7)$$

The  $a = 0$  term means  $k+1$ . The equality follows by expanding  $(1-q^j)^r$  binomially and summing the geometric powers  $q^{aj}$ . Equations (7.4)–(7.5) and the Bell formulae give

$$S_q(n, k) = \frac{1}{(n-k)!} \mathcal{B}_{n-k} \left( ((r-1)! U_r(k))_{r=1}^{n-k} \right), \quad (7.8)$$

$$s_q(n, k) = \frac{(-1)^{n-k}}{(n-k)!} \mathcal{B}_{n-k} \left( ((-1)^{r-1} (r-1)! U_r(n-1))_{r=1}^{n-k} \right). \quad (7.9)$$

These are recurrence-free finite formulae involving only binomial coefficients,  $q$ -integers, factorials, and the finite sum (1.4). For example,

$$S_q(n, n-1) = \sum_{j=1}^{n-1} [j]_q = \sum_{a=0}^{n-2} (n-1-a) q^a,$$

and

$$S_q(n, n-2) = \frac{1}{2} (U_1(n-2)^2 + U_2(n-2)).$$

The latter uses the alphabet ending at  $[n-2]_q$ , not at  $[n-1]_q$ .

## 7.4 The normal-ordering normalization

Define

$$\tilde{S}_q(n, k) = q^{\binom{k}{2}} S_q(n, k). \quad (7.10)$$

Then

$$\tilde{S}_q(n+1, k) = [k]_q \tilde{S}_q(n, k) + q^{k-1} \tilde{S}_q(n, k-1). \quad (7.11)$$

Both this and (7.3) occur in  $q$ -Stirling literature; the restricted-growth-word viewpoint and related identities are discussed in [5]. The defining recurrence must accompany any comparison of conventions.

**Theorem 7.3** (Normal ordering and factorial moments). *On polynomials,*

$$(XD_q)^n = \sum_{k=0}^n \tilde{S}_q(n, k) X^k D_q^k. \quad (7.12)$$

For integers  $m, n \geq 0$ ,

$$[m]_q^n = \sum_{k=0}^{\min(m, n)} \tilde{S}_q(n, k) [k]_q! \begin{bmatrix} m \\ k \end{bmatrix}_q. \quad (7.13)$$

*Proof.* Induction from (6.2) gives  $D_q X^k = [k]_q X^{k-1} + q^k X^k D_q$ . Left multiplication of the right side of (7.12) by  $XD_q$  therefore produces recurrence (7.11), with the correct initial value. This proves normal ordering. Apply both sides to  $x^m$ . The left side is  $[m]_q^n x^m$ , and  $X^k D_q^k x^m = [k]_q! \begin{bmatrix} m \\ k \end{bmatrix}_q x^m$ ; terms with  $k > m$  vanish. Comparison proves (7.13).  $\square$

For  $n = 2$ , normal ordering reads  $(XD_q)^2 = XD_q + qX^2D_q^2$ . Thus  $\tilde{S}_q(2, 2) = q$ , while  $S_q(2, 2) = 1$ . This tiny example is an effective audit of the two normalizations.



## Chapter 8

# Weighted Bell polynomials and Riordan composition

### 8.1 Gaussian convolution as a change of coefficient coordinates

Set  $w_n = [n]_q!$ , and record a sequence by

$$A(z) = \sum_{n \geq 0} a_n \frac{z^n}{w_n}.$$

Ordinary multiplication of series is then the Gaussian convolution

$$(a \star_q b)_n = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q a_k b_{n-k}. \quad (8.1)$$

Associativity and commutativity follow from ordinary series multiplication. More explicitly, for  $s$  factors the convolution coefficient is

$$\sum_{i_1 + \dots + i_s = n} \frac{[n]_q!}{[i_1]_q! \dots [i_s]_q!} a_{i_1}^{(1)} \dots a_{i_s}^{(s)}. \quad (8.2)$$

The factorial ratio is the Gaussian multinomial. It is polynomial because it is the product  $\begin{bmatrix} n \\ i_1 \end{bmatrix}_q \begin{bmatrix} n - i_1 \\ i_2 \end{bmatrix}_q \dots$ . Thus associativity extends polynomially to every specialization of  $q$ , even when the divided-power representation itself is not available.

**Proposition 8.1** (Explicit convolution inverse). *If  $a_0$  is invertible, the inverse  $b$  for  $\star_q$  satisfies  $b_0 = a_0^{-1}$ , and for  $n > 0$ ,*

$$b_n = \frac{1}{a_0} \sum_{r=1}^n \frac{(-1)^r}{a_0^r} \sum_{\substack{i_1 + \dots + i_r = n \\ i_j \geq 1}} \frac{[n]_q!}{[i_1]_q! \dots [i_r]_q!} \prod_{j=1}^r a_{i_j}. \quad (8.3)$$

*Proof.* Write  $A(z) = a_0(1 + U(z))$ , where  $U(0) = 0$ . The formal geometric expansion  $A(z)^{-1} = a_0^{-1} \sum_{r \geq 0} (-U(z))^r$  is coefficientwise finite. Multiplying its degree- $n$  coefficient by  $w_n$  gives the displayed formula.  $\square$

In particular, with

$$e_q(z) = \sum_{n \geq 0} \frac{z^n}{[n]_q!}, \quad E_q(z) = \sum_{n \geq 0} q^{\binom{n}{2}} \frac{z^n}{[n]_q!},$$

Gaussian inversion is the coefficient identity  $e_q(z)E_q(-z) = 1$ .

## 8.2 Composition coefficients without an ambiguous “ $q$ -Bell” convention

Let  $B(z) = \sum_{j \geq 1} b_j z^j / w_j$ . Define

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n}{w_k} [z^n] B(z)^k. \quad (8.4)$$

This is a specified weighted coefficient family; it is not a claim that all objects called  $q$ -Bell polynomials in the literature have this meaning.

**Theorem 8.2.** For  $0 \leq k \leq n$ ,

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n k!}{w_k} \sum_{\substack{m_1, \dots, m_n \geq 0 \\ \sum m_j = k, \sum j m_j = n}} \prod_{j=1}^n \frac{1}{m_j!} \left( \frac{b_j}{w_j} \right)^{m_j}. \quad (8.5)$$

Equivalently,

$$\mathfrak{B}_{n,k}^{[q]}(b) = \frac{w_n k!}{w_k n!} \mathcal{B}_{n,k} \left( \frac{1! b_1}{w_1}, \frac{2! b_2}{w_2}, \dots \right). \quad (8.6)$$

If  $A(z) = \sum_{k \geq 0} a_k z^k / w_k$ , then

$$w_n [z^n] A(B(z)) = \sum_{k=0}^n a_k \mathfrak{B}_{n,k}^{[q]}(b). \quad (8.7)$$

*Proof.* Expand  $B(z)^k$  by the ordinary multinomial theorem. A multiplicity vector  $(m_j)$  contributes  $k! \prod_j (b_j / w_j)^{m_j} / m_j!$ . The two constraints count the factors and their total degree. This proves (8.5). Compare it with the defining finite sum for  $\mathcal{B}_{n,k}$  to obtain (8.6). Finally, expand  $A(B)$  in powers of  $B$  and use (8.4).  $\square$

At  $q = 1$ ,  $w_n = n!$ , and  $\mathfrak{B}_{n,k}^{[q]}$  reduces to the ordinary exponential Bell polynomial  $\mathcal{B}_{n,k}$ . At  $q = 0$ ,  $w_n = 1$ , and it becomes the ordinary coefficient  $[z^n] B(z)^k$ . This cleanly separates coefficient normalization from the choice of a deformation of set partitions.

## 8.3 Weighted Riordan arrays

For  $A(0) \neq 0$  and  $B(0) = 0$  with invertible linear coefficient, put

$$\mathcal{M}_q(A, B)_{n,k} = \frac{w_n}{w_k} [z^n] A(z) B(z)^k. \quad (8.8)$$

**Theorem 8.3.** These lower triangular matrices obey

$$\mathcal{M}_q(A, B) \mathcal{M}_q(C, D) = \mathcal{M}_q(A C(B), D(B)), \quad (8.9)$$

$$\mathcal{M}_q(A, B)^{-1} = \mathcal{M}_q \left( \frac{1}{A(\overline{B})}, \overline{B} \right), \quad (8.10)$$

where  $\overline{B}$  is the ordinary compositional inverse of  $B$ .

*Proof.* The  $(n, k)$  entry of the product equals

$$\frac{w_n}{w_k} \sum_j [z^n] A(z) B(z)^j [t^j] C(t) D(t)^k.$$

For fixed  $n$ , the sum is finite because  $B(0) = 0$ . Summing before taking  $[z^n]$  produces  $A(z) C(B(z)) D(B(z))^k$ , proving the product law. Inserting  $D = \overline{B}$  and  $C = 1/A(\overline{B})$  gives  $\mathcal{M}_q(1, z)$ , the identity matrix. The same substitution in the other order gives the other inverse identity.  $\square$

This is diagonal transport of ordinary Riordan calculus. It extends the source's Riordan framework while making clear which operation is unchanged: composition is still ordinary substitution of formal power series.

## 8.4 Why Jackson composition is a different problem

The first-order chain rule for the Jackson derivative is

$$D_q(F \circ G)(x) = F[G(x), G(qx)] D_q G(x), \quad (8.11)$$

where  $F[u, v] = (F(u) - F(v))/(u - v)$ , with its polynomial continuation on the diagonal. This follows simply by factoring the numerator in (6.1). It is not usually  $(D_q F)(G(x)) D_q G(x)$ . For example, with  $F(u) = u^2$  and  $G(x) = x^2$ , the actual answer is  $[4]_q x^3$ , whereas that guess gives  $[2]_q^2 x^3$ . The weighted Bell family in this chapter describes ordinary composition and must not be silently substituted into a higher Jackson chain rule.

## Chapter 9

# Lagrange inversion with geometric product data

### 9.1 The ordinary inversion theorem over a $q$ -dependent field

**Theorem 9.1** (Formal Lagrange–Bürmann). *Let  $\Phi(0) = 1$ . There is a unique  $y(z) \in zK[[z]]$  satisfying  $y = z\Phi(y)$ . For  $n \geq 1$ ,*

$$[z^n]H(y(z)) = \frac{1}{n}[t^{n-1}]H'(t)\Phi(t)^n. \quad (9.1)$$

*In particular, for  $s \geq 1$  and  $n \geq s$ ,*

$$[z^n]y(z)^s = \frac{s}{n}[t^{n-s}]\Phi(t)^n. \quad (9.2)$$

*Proof.* The equation determines the coefficient of degree  $n$  of  $y$  from lower degrees, starting with  $[z]y = 1$ . This proves existence and uniqueness. The inverse substitution is  $z = t/\Phi(t)$ . For a Laurent series, residue means the coefficient of degree  $-1$ . The residue of a formal derivative is zero, and the substitution rule  $\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du$  follows on monomials: for powers other than  $-1$ , the integrand is a derivative; for the power  $-1$ , the logarithmic derivative has residue one. Thus

$$\begin{aligned} [z^n]H(y(z)) &= \text{Res}_t H(t) \left( \frac{\Phi(t)^n}{t^{n+1}} - \frac{\Phi(t)^{n-1}\Phi'(t)}{t^n} \right) dt \\ &= \frac{1}{n} \text{Res}_t H'(t) \frac{\Phi(t)^n}{t^n} dt. \end{aligned}$$

The second equality is obtained by taking the zero residue of the derivative of  $H(t)\Phi(t)^n t^{-n}$ . It is (9.1); taking  $H(t) = t^s$  proves (9.2).  $\square$

This is the ordinary theorem used in the source, now over a field containing  $q$ ; see also the proof-oriented survey [8].

### 9.2 A recurrence-free inverse for a product of shifted factorials

**Theorem 9.2.** *Suppose  $\Phi(t)$  is the product (4.8), with logarithmic power sums  $P_r$ . The solution of  $y = z\Phi(y)$  has coefficients*

$$[z^n]y(z) = \frac{1}{n!} \mathcal{B}_{n-1} \left( (n(r-1)!P_r)_{r=1}^{n-1} \right) \quad (n \geq 1). \quad (9.3)$$

*More generally,*

$$[z^n]y(z)^s = \frac{s}{n(n-s)!} \mathcal{B}_{n-s} \left( (n(r-1)!P_r)_{r=1}^{n-s} \right), \quad n \geq s \geq 1. \quad (9.4)$$

*Proof.* The logarithm of  $\Phi(t)^n$  has power sums  $nP_r$ . Apply Theorem 4.2 inside (9.2). The case  $s = 1$  has denominator  $n(n-1)! = n!$ .  $\square$

This solves, coefficient by coefficient and without auxiliary recurrence, the inverse problem for

$$f(x) = x \prod_{\ell=1}^L (a_\ell x; q^{d_\ell})_{N_\ell}^{\alpha_\ell}.$$

For  $f(x) = x(x; q)_M^\alpha$ , the formula becomes

$$[z^n]f^{-1}(z) = \frac{1}{n!} \mathcal{B}_{n-1} \left( (\alpha n(r-1)! [M]_{q^r})_{r=1}^{n-1} \right). \quad (9.5)$$

An alternative finite sum, obtained by expanding each factor separately, is

$$[z^n]f^{-1}(z) = \frac{1}{n} \sum_{\substack{r_0, \dots, r_{M-1} \geq 0 \\ \sum r_j = n-1}} \prod_{j=0}^{M-1} \frac{\alpha n^{\overline{r_j}}}{r_j!} q^{\sum j r_j}. \quad (9.6)$$

Here the rising factorial makes the formula polynomial in  $\alpha$ , including values at which gamma-quotient representations would be inconvenient.

**Example 9.3** (A two-color inverse). The inverse of  $x(1-x)(1-qx)$  begins

$$\begin{aligned} y(z) = & z + (1+q)z^2 + (2+3q+2q^2)z^3 \\ & + (5+10q+10q^2+5q^3)z^4 + O(z^5). \end{aligned}$$

At  $q = 1$ , the coefficient formula is  $n^{-1} \binom{3n-2}{n-1}$ , giving 1, 2, 7, 30, ...

The coefficients for positive integral  $\alpha$  count rooted plane trees whose children are split into  $\alpha M$  ordered blocks; a child in a block of color  $j$  has weight  $q^j$ . Each vertex contributes one factor  $\Phi(t)$ , and the root contributes  $z$ , proving the functional equation. Every tree with  $n$  vertices has  $n-1$  edges. Reversing all color labels  $j \mapsto M-1-j$  therefore proves

$$[z^n]y(z; q^{-1}) = q^{-(M-1)(n-1)} [z^n]y(z; q). \quad (9.7)$$

The same identity follows algebraically by reversing the finite product and using homogeneity in (9.5). The tree argument additionally proves nonnegativity and integrality in this parameter range.

#### Ordinary inversion is not obtained by replacing $n$ with $[n]_q$

The factor  $1/n$  in Lagrange inversion is an ordinary integer factor. For  $M = 1$ , the equation is  $y = z/(1-y)$ , independent of  $q$ , and its coefficients are Catalan numbers. Replacing  $1/n$  by  $1/[n]_q$  would make the answer depend on  $q$ , contradicting the defining equation. Genuine  $q$ -shifted inversion theories change additional algebraic structure; they are not the operation performed here.

### 9.3 A two-variable inverse with its Jacobian factor exposed

The source also develops multivariate Good inversion. The following complete specialization shows how geometric-product Bell data enter a coupled system, while retaining the correction that a naive product of one-variable formulae would omit.

**Theorem 9.4** (Cross-coupled inversion). *Let  $A(0) = B(0) = 1$ , and let*

$$u = z_1 A(v), \quad v = z_2 B(u).$$

*For  $m, n > 0$ ,  $r, s \geq 0$ ,  $m \geq r$ , and  $n \geq s$ ,*

$$[z_1^m z_2^n] u^r v^s = \frac{ms + nr - rs}{mn} [t^{n-s}] A(t)^m [t^{m-r}] B(t)^n. \quad (9.8)$$

*Each coefficient on the right has the Bell form (4.9) when  $A, B$  are finite geometric products.*

*Proof.* Substitute the first equation into the second:  $v = z_2 B(z_1 A(v))$ . Apply the one-variable formula (9.1) in  $z_2$ , with  $z_1$  a parameter, to  $u^r v^s = z_1^r A(v)^r v^s$ . The derivative is  $st^{s-1} A(t)^r + rt^s A(t)^{r-1} A'(t)$ . Furthermore,

$$[z_1^{m-r}] B(z_1 A(t))^n = A(t)^{m-r} [w^{m-r}] B(w)^n.$$

The first derivative term therefore contributes  $\frac{s}{n} [t^{n-s}] A(t)^m$  times the  $B$ -coefficient. The second contributes

$$\frac{r}{n} [t^{n-s-1}] A(t)^{m-1} A'(t) = \frac{r(n-s)}{mn} [t^{n-s}] A(t)^m.$$

Adding gives  $(ms + nr - rs)/(mn)$ . This argument also covers  $r = 0$  or  $s = 0$ , since the corresponding derivative term is zero.  $\square$

For comparison, the determinant in the two-variable Good formula is

$$1 - \frac{t_2 A'(t_2)}{A(t_2)} \frac{t_1 B'(t_1)}{B(t_1)}.$$

Its correction subtracts  $(n-s)(m-r)/(mn)$  from one, which is exactly the prefactor in (9.8). This gives an independent check on the indices and on the otherwise easily omitted Jacobian contribution.

## Chapter 10

# The coefficients of a Gaussian polynomial

### 10.1 A divisor-sum formula for each coefficient

Write  $m = n - k$ ,  $D = k(n - k)$ , and

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{r=0}^D c_{n,k}(r) q^r.$$

This chapter extracts coefficients in the *base*  $q$ , rather than in an auxiliary product variable. Since  $G_{n,k}(0) = 1$ , it has a formal logarithm in  $\mathbb{Q}[[q]]$ .

**Theorem 10.1.** *For  $r \geq 1$ , define the finite divisor sum*

$$A_r(n, k) = \sum_{\substack{d|r \\ 1 \leq d \leq k}} d - \sum_{\substack{d|r \\ m < d \leq n}} d. \quad (10.1)$$

*Then*

$$c_{n,k}(r) = \frac{1}{r!} \mathcal{B}_r \left( ((j-1)! A_j(n, k))_{j=1}^r \right), \quad (10.2)$$

*or, with no named auxiliary polynomial,*

$$c_{n,k}(r) = \sum_{\substack{v_1, \dots, v_r \geq 0 \\ \sum j v_j = r}} \prod_{j=1}^r \frac{A_j(n, k)^{v_j}}{j^{v_j} v_j!}. \quad (10.3)$$

*Also  $c_{n,k}(0) = 1$ , and the formula gives zero for  $r > D$ .*

*Proof.* The finite product quotient gives

$$\begin{aligned} \log G_{n,k}(q) &= \sum_{d=1}^k \sum_{a \geq 1} \frac{q^{da}}{a} - \sum_{d=m+1}^n \sum_{a \geq 1} \frac{q^{da}}{a} \\ &= \sum_{r \geq 1} \frac{A_r(n, k)}{r} q^r. \end{aligned}$$

For fixed  $r$ , only divisors  $d \mid r$  contribute, with coefficient  $1/(r/d) = d/r$ . The Bell exponential formula proves the result. The original series is a polynomial of degree  $D$ , proving the last assertion.  $\square$

Although individual terms in (10.3) are rational and can have signs, the answer is a nonnegative integer by the subset definition. For example, when  $(n, k) = (4, 2)$ , the first kernels are  $A_1 = 1, A_2 = 3, A_3 = -2, A_4 = -1$ . They yield

$$c(1) = 1, \quad c(2) = \frac{1+3}{2} = 2, \quad c(3) = \frac{1+9-4}{6} = 1,$$

and  $c(4) = 1$ , as expected.

## 10.2 A second finite formula by inclusion–exclusion

**Proposition 10.2.** *For  $r \geq 0$ ,*

$$c_{n,k}(r) = \sum_{\varepsilon \in \{0,1\}^k} (-1)^{\sum \varepsilon_j} \sum_{\substack{v_1, \dots, v_k \geq 0 \\ \sum j v_j = r - \sum (m+j) \varepsilon_j}} 1. \quad (10.4)$$

*An impossible inner constraint contributes zero. All inner variables may be bounded by  $v_j \leq \lfloor r/j \rfloor$ .*

*Proof.* Expand each numerator factor  $1 - q^{m+j}$  by its two terms and each reciprocal factor  $(1 - q^j)^{-1}$  as a geometric series. The sign records which numerator terms were selected, and the displayed constraint is precisely the total exponent. Coefficient extraction proves the formula.  $\square$

The two finite formulae organize the same coefficient differently. The first collects repeated divisibility contributions into Bell blocks; the second uses restricted partitions and inclusion–exclusion. Neither needs earlier values of  $c_{n,k}(r)$ .

## 10.3 Finite Fourier extraction

Let  $L > D$ , and let  $\omega = e^{2\pi i/L}$ . Orthogonality of roots of unity implies, for  $0 \leq r \leq D$ ,

$$c_{n,k}(r) = \frac{1}{L} \sum_{j=0}^{L-1} \omega^{-rj} G_{n,k}(\omega^j). \quad (10.5)$$

Indeed, expand the polynomial and use  $L^{-1} \sum_{j=0}^{L-1} \omega^{j(s-r)} = \delta_{sr}$  in this degree range. Without the condition  $L > D$ , the same expression gives the sum of coefficients with exponent congruent to  $r \pmod{L}$ . Later,  $q$ -Lucas supplies a direct root-of-unity evaluation of every summand.

This identity is exact in a cyclotomic field. Floating-point use can suffer from cancellation, so it is not automatically a preferable numerical algorithm. For exact coefficient production, polynomial Pascal recursion or the divisor-sum formula is usually simpler.



## Chapter 11

# Parameter derivatives and the expansion at one

### 11.1 Why the logarithmic parameter is the natural coordinate

Put  $m = n - k$ ,  $D = km$ , and write

$$G_{n,k}(q) = \begin{bmatrix} n \\ k \end{bmatrix}_q = \sum_{j=0}^D c_{n,k}(j) q^j.$$

The substitution  $q = e^t$  converts powers of  $q$  into exponentials. Thus ordinary derivatives in  $t$  give power moments of the coefficient sequence, whereas derivatives in  $q$  give falling-factorial moments. This is precisely the distinction for which the source article uses the Stirling basis changes. Let  $\vartheta = q \, d/dq$ , so that  $\frac{d}{dt} F(e^t) = (\vartheta F)(e^t)$ .

**Lemma 11.1** (The regular logarithm of an exponential difference). *As a formal series at  $u = 0$ ,*

$$\log \frac{e^u - 1}{u} = \frac{u}{2} + \sum_{r \geq 2} \frac{\beta_r}{r} \frac{u^r}{r!}. \quad (11.1)$$

*All odd coefficients after the linear one vanish.*

*Proof.* The quotient inside the logarithm has constant term one. Its logarithmic derivative is

$$\frac{e^u}{e^u - 1} - \frac{1}{u} = 1 + \frac{1}{u} \left( \frac{u}{e^u - 1} - 1 \right) = \frac{1}{2} + \sum_{r \geq 2} \beta_r \frac{u^{r-1}}{r!}.$$

Integrate formally and use the zero constant term of the logarithm. The function  $u/(e^u - 1) + u/2$  is even, as direct substitution of  $-u$  shows. Hence  $\beta_{2j+1} = 0$  for  $j \geq 1$ .  $\square$

**Theorem 11.2** (All-order Gaussian cumulants). *Define*

$$\kappa_1(n, k) = \frac{km}{2}, \quad \kappa_r(n, k) = \frac{\beta_r}{r} \sum_{j=1}^k ((m+j)^r - j^r) \quad (r \geq 2). \quad (11.2)$$

*Then*

$$\log \frac{G_{n,k}(e^t)}{\binom{n}{k}} = \sum_{r \geq 1} \kappa_r(n, k) \frac{t^r}{r!}, \quad (11.3)$$

$$G_{n,k}(e^t) = \binom{n}{k} \sum_{r \geq 0} \mathcal{B}_r(\kappa_1, \dots, \kappa_r) \frac{t^r}{r!}. \quad (11.4)$$

These are both formal identities and convergent local Taylor expansions. In particular,

$$\kappa_2 = \frac{D(n+1)}{12}, \quad \kappa_3 = 0, \quad \kappa_4 = -\frac{D(n+1)(n(n+1)-D)}{120}. \quad (11.5)$$

*Proof.* The cancelled product representation is

$$\frac{G_{n,k}(e^t)}{\binom{n}{k}} = \prod_{j=1}^k \frac{(e^{(m+j)t} - 1)/((m+j)t)}{(e^{jt} - 1)/(jt)}.$$

Every factor now has constant term one. Apply Theorem 11.1 to each numerator and denominator. The linear coefficient is  $\frac{1}{2} \sum_{j=1}^k m = km/2$ ; the higher coefficients are exactly (11.2). Exponentiation and the Bell generating function give (11.4).

For the second coefficient,  $\sum_{j=1}^k ((m+j)^2 - j^2) = km(n+1)$ . For the fourth, expansion of the fourth powers and summation of powers of  $j$  gives

$$\sum_{j=1}^k ((m+j)^4 - j^4) = km(n+1)(n(n+1) - km).$$

Together with  $\beta_2 = 1/6$ ,  $\beta_3 = 0$ , and  $\beta_4 = -1/30$ , these are (11.5). The local analytic assertions follow because the cancelled product is analytic and nonzero near  $t = 0$ .  $\square$

### 11.1.1 Finite Bernoulli-polynomial form

No power-sum recurrence is necessary in (11.2). Define  $\beta_r(x) = \sum_{j=0}^r \binom{r}{j} \beta_j x^{r-j}$ . Multiplying its generating function  $te^{xt}/(e^t - 1)$  by  $e^t - 1$  shows  $\beta_{r+1}(x+1) - \beta_{r+1}(x) = (r+1)x^r$ . Consequently

$$P_r(M) := \sum_{j=1}^M j^r = \frac{\beta_{r+1}(M+1) - \beta_{r+1}(1)}{r+1}.$$

The sum in (11.2) is  $P_r(n) - P_r(m) - P_r(k)$ . This also makes its symmetry in  $k, m$  manifest.

## 11.2 Exact coefficient moments and probability

**Corollary 11.3** (Power and factorial moments). *For  $r \geq 0$ ,*

$$\sum_{j=0}^D j^r c_{n,k}(j) = \binom{n}{k} \mathcal{B}_r(\kappa_1, \dots, \kappa_r), \quad (11.6)$$

$$G_{n,k}^{(r)}(1) = \binom{n}{k} \sum_{a=0}^r s(r, a) \mathcal{B}_a(\kappa_1, \dots, \kappa_a). \quad (11.7)$$

For  $r = 0$ , the first sum is the coefficient sum, with  $0^0 = 1$ .

*Proof.* Differentiate the finite sum  $G_{n,k}(e^t) = \sum_j c_{n,k}(j) e^{jt}$ . This proves (11.6). The identity  $j^r = \sum_a s(r, a) j^a$ , which defines the signed Stirling coefficients, gives (11.7).  $\square$

Choosing a partition uniformly from the partitions in the  $k$ -by- $m$  rectangle gives a random area  $X$  with  $\Pr(X = j) = c_{n,k}(j)/\binom{n}{k}$ . Its moment generating function is the left side of (11.4), divided by  $\binom{n}{k}$ . Thus the  $\kappa_r$  are its cumulants, and

$$\mathbb{E}X = \frac{D}{2}, \quad \text{Var}(X) = \frac{D(n+1)}{12}. \quad (11.8)$$

Complementing the rectangle sends  $X$  to  $D - X$ ; hence every odd central moment is zero. Every even central moment is given by  $\mathcal{B}_{2r}(0, \kappa_2, 0, \kappa_4, \dots, \kappa_{2r})$ . These assertions concern a finite distribution and require no limiting probabilistic theorem.

**Example 11.4.** For  $G_{4,2}(q) = 1 + q + 2q^2 + q^3 + q^4$ , the coefficient sum is six,  $\kappa_1 = 2$ ,  $\kappa_2 = 5/3$ , and  $\kappa_4 = -8/3$ . Therefore

$$G'_{4,2}(1) = 6 \cdot 2 = 12, \quad (\vartheta^2 G_{4,2})(1) = 6(2^2 + 5/3) = 34,$$

whereas  $G''_{4,2}(1) = 34 - 12 = 22$ . Confusing the two kinds of differentiation would already give an incorrect second derivative.

### 11.3 Arbitrary parameter points and Eulerian rational functions

For  $s \geq 0$ , define the rational function

$$\text{Li}_{-s}(u) = \left( u \frac{d}{du} \right)^s \frac{u}{1-u}.$$

For  $|u| < 1$  this is  $\sum_{j \geq 1} j^s u^j$ , but the rational expression is the definition needed outside that disk. Set  $A_0(u) = 1$ . Then

$$\text{Li}_{-s}(u) = \frac{u A_s(u)}{(1-u)^{s+1}}, \quad A_{s+1}(u) = (1+su)A_s(u) + u(1-u)A'_s(u). \quad (11.9)$$

The formula follows by applying  $u d/du$  to the fraction. For  $s \geq 1$ , the coefficients of  $A_s$  are the ordinary Eulerian numbers indexed by the number of descents: insertion of the largest letter gives the same coefficient recurrence and initial value. Alternatively, (11.9) is an independent algebraic definition of this Eulerian normalization.

**Theorem 11.5** (Logarithmic parameter derivatives). *At a nonzero point  $q$  for which none of  $q^1, \dots, q^n$  equals one, define*

$$L_r(q) = \sum_{j=1}^k (j^r \text{Li}_{1-r}(q^j) - (m+j)^r \text{Li}_{1-r}(q^{m+j})). \quad (11.10)$$

Then, for  $r \geq 0$ ,

$$\vartheta^r G_{n,k}(q) = G_{n,k}(q) \mathcal{B}_r(L_1(q), \dots, L_r(q)), \quad (11.11)$$

$$G_{n,k}^{(r)}(q) = q^{-r} G_{n,k}(q) \sum_{a=0}^r s(r, a) \mathcal{B}_a(L_1(q), \dots, L_a(q)). \quad (11.12)$$

All  $L_r$  are explicit rational functions, by (11.9).

*Proof.* For  $a \geq 1$ , repeated logarithmic differentiation gives

$$\vartheta^r \log(1 - q^a) = -a^r \text{Li}_{1-r}(q^a), \quad r \geq 1.$$

Subtracting denominator contributions from numerator contributions gives  $\vartheta^r \log G = L_r$ . In the local coordinate  $qe^t$ , the logarithm of  $G(qe^t)/G(q)$  is  $\sum_{r \geq 1} L_r(q) t^r / r!$ . Exponentiate and use Bell polynomials. Finally, the operator identity

$$q^r \frac{d^r}{dq^r} = \vartheta(\vartheta - 1) \cdots (\vartheta - r + 1) = \sum_{a=0}^r s(r, a) \vartheta^a \quad (11.13)$$

follows by applying both sides to each monomial. It proves the second formula.  $\square$

The displayed domain avoids unnecessary cancellation inside individual rational terms. At roots of unity, the Gaussian polynomial is still regular, but some  $L_r$  may have poles. The correct replacement is a deflated local factorization, not termwise substitution into (11.10).

## Chapter 12

# Cyclotomic structure and all-order root-of-unity jets

### 12.1 Cyclotomic factorization and carries

Let  $\Phi_d(q)$  denote the monic polynomial whose roots are the primitive  $d$ -th roots of unity. The partition of roots by their exact order gives  $q^n - 1 = \prod_{d|n} \Phi_d(q)$ .

**Theorem 12.1** (Squarefree cyclotomic factorization). *For  $0 \leq k \leq n$ ,  $m = n - k$ ,*

$$G_{n,k}(q) = \prod_{d=2}^n \Phi_d(q)^{\epsilon_d}, \quad \epsilon_d = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{m}{d} \right\rfloor \in \{0, 1\}. \quad (12.1)$$

*Consequently every nonconstant Gaussian polynomial has distinct roots, all on the unit circle. Its degree gives the identity*

$$\sum_{d=2}^n \varphi(d) \epsilon_d = k(n - k). \quad (12.2)$$

*Proof.* Factor each  $1 - q^a$  in the factorial quotient. There are  $\lfloor n/d \rfloor$  multiples of  $d$  in its numerator factorial, and the denominator subtracts the corresponding counts for  $k$  and  $m$ . The exponent of  $\Phi_1 = q - 1$  is  $n - k - m = 0$ ; the signs also cancel. Write  $k = ud + v$ ,  $m = wd + x$ , with  $0 \leq v, x < d$ . Then  $\epsilon_d = \lfloor (v + x)/d \rfloor$ , which is either zero or one. The roots of different cyclotomic polynomials are disjoint, and each is simple. Finally  $\deg \Phi_d = \varphi(d)$ , and degrees add in the product.  $\square$

This carry interpretation is particularly useful: vanishing at a primitive  $d$ -th root occurs precisely when the residues of  $k$  and  $n - k$  produce a carry upon addition modulo  $d$ .

### 12.2 The finite $q$ -Lucas theorem

The following classical root-of-unity formula is often called  $q$ -Lucas; see, for example, [7]. Its proof is short enough to include, and the phase bookkeeping is important.

**Theorem 12.2** ( $q$ -Lucas). *Let  $\zeta$  be a primitive  $d$ -th root of unity, with  $d \geq 1$ . Write  $n = ad + b$ ,  $k = rd + s$ , where  $0 \leq b, s < d$ . Then*

$$G_{n,k}(\zeta) = \binom{a}{r} G_{b,s}(\zeta). \quad (12.3)$$

*The right side is zero when  $s > b$ .*

*Proof.* The finite product theorem and a division into full blocks give

$$\prod_{j=0}^{n-1} (1 + \zeta^j z) = (1 - (-z)^d)^a \prod_{j=0}^{b-1} (1 + \zeta^j z).$$

The second factor has degree smaller than  $d$ . Hence its only possible contribution to  $z^{rd+s}$  has degree  $s$ , and coefficient comparison therefore gives

$$\zeta^{\binom{k}{2}} G_{n,k}(\zeta) = (-1)^{r(d+1)} \binom{a}{r} \zeta^{\binom{s}{2}} G_{b,s}(\zeta).$$

The phase identity  $\zeta^{\binom{rd+s}{2} - \binom{s}{2}} = (-1)^{r(d+1)}$  finishes the proof. For odd  $d$ , the exponent is a multiple of  $d$ . For even  $d$ , it is congruent to  $rd/2$  modulo  $d$ , giving  $(-1)^r$ . These are exactly the two values on the right. The same argument covers  $d = 1$ .  $\square$

For example,

$$G_{2a,2r}(-1) = \binom{a}{r}, \quad G_{2a,2r+1}(-1) = 0,$$

while  $G_{2a+1,2r}(-1) = G_{2a+1,2r+1}(-1) = \binom{a}{r}$ . If  $d \mid n$ , (12.3) is also a fixed-subset count: a permutation consisting of  $n/d$  disjoint  $d$ -cycles fixes a  $k$ -element subset exactly when the subset is a union of whole cycles. There are  $\binom{n/d}{k/d}$  such subsets if  $d \mid k$ , and none otherwise. This provides a concrete combinatorial interpretation for these root evaluations.

### 12.3 Deflating all apparent singularities

A value formula such as  $q$ -Lucas does not determine derivatives at the root. The next theorem supplies every derivative, including the case where the Gaussian polynomial itself vanishes. The calculation uses the logarithmic coordinate  $q = \zeta e^t$  and removes powers of  $t$  before taking a logarithm.

For an integer  $a \geq 1$ , define

$$c_a(\zeta) = \begin{cases} -a, & d \mid a, \\ 1 - \zeta^a, & d \nmid a. \end{cases} \quad (12.4)$$

For  $r \geq 1$ , define

$$\eta_r(a; \zeta) = \begin{cases} a/2, & d \mid a, r = 1, \\ \beta_r a^r / r, & d \mid a, r \geq 2, \\ -a^r \operatorname{Li}_{1-r}(\zeta^a), & d \nmid a. \end{cases} \quad (12.5)$$

All rational functions in the last line are evaluated away from their pole at one. Put

$$\nu = \left\lfloor \frac{n}{d} \right\rfloor - \left\lfloor \frac{k}{d} \right\rfloor - \left\lfloor \frac{m}{d} \right\rfloor, \quad (12.6)$$

$$C_\zeta = \prod_{j=1}^k \frac{c_{m+j}(\zeta)}{c_j(\zeta)}, \quad (12.7)$$

$$\lambda_r = \sum_{j=1}^k (\eta_r(m+j; \zeta) - \eta_r(j; \zeta)). \quad (12.8)$$

Here  $\nu = 0$  when  $d = 1$ , and always  $C_\zeta \neq 0$ .

**Theorem 12.3** (Uniform all-order local formula). *With the definitions above,*

$$G_{n,k}(\zeta e^t) = C_\zeta t^\nu \exp \left( \sum_{r \geq 1} \lambda_r \frac{t^r}{r!} \right). \quad (12.9)$$

In particular, for  $r \geq 0$ ,

$$[t^{\nu+r}]G_{n,k}(\zeta e^t) = \frac{C_\zeta}{r!} \mathcal{B}_r(\lambda_1, \dots, \lambda_r). \quad (12.10)$$

Every ordinary Taylor derivative at  $\zeta$  is the finite sum

$$G_{n,k}^{(R)}(\zeta) = C_\zeta \zeta^{-R} \sum_{j=\nu}^R s(R, j) \frac{j!}{(j-\nu)!} \mathcal{B}_{j-\nu}(\lambda_1, \dots, \lambda_{j-\nu}). \quad (12.11)$$

An empty sum is zero. Formula (12.9) holds formally in  $\mathbb{Q}(\zeta)[[t]]$  and analytically near  $t = 0$ .

*Proof.* For each factor in the product quotient, consider two cases. If  $d \mid a$ , then

$$1 - \zeta^a e^{at} = -at \frac{e^{at} - 1}{at} = c_a(\zeta) t \exp \left( \frac{at}{2} + \sum_{r \geq 2} \frac{\beta_r a^r}{r} \frac{t^r}{r!} \right)$$

by Theorem 11.1. If  $d \nmid a$ , the factor has nonzero constant term  $c_a(\zeta)$ , and repeated differentiation yields

$$\log \frac{1 - \zeta^a e^{at}}{1 - \zeta^a} = - \sum_{r \geq 1} a^r \text{Li}_{1-r}(\zeta^a) \frac{t^r}{r!}.$$

Multiplying numerator factors and dividing denominator factors produces the constant  $C_\zeta$ , the net power  $t^\nu$ , and the sum of logarithms  $\sum \lambda_r t^r / r!$ . This proves (12.9); Bell expansion proves (12.10).

The operator  $\vartheta$  becomes  $d/dt$ . Therefore

$$(\vartheta^j G)(\zeta) = \begin{cases} 0, & j < \nu, \\ C_\zeta \frac{j!}{(j-\nu)!} \mathcal{B}_{j-\nu}(\lambda_1, \dots, \lambda_{j-\nu}), & j \geq \nu. \end{cases}$$

Apply (11.13) at  $q = \zeta$  to get (12.11). Finally, after deflation all factors are analytic units near zero, so their normalized logarithms converge locally. No logarithm of a vanishing quantity was used.  $\square$

**Corollary 12.4** (First two derivatives). *At a nonzero value,  $\nu = 0$ ,*

$$G(\zeta) = C_\zeta, \quad G'(\zeta) = C_\zeta \zeta^{-1} \lambda_1, \quad G''(\zeta) = C_\zeta \zeta^{-2} (\lambda_1^2 + \lambda_2 - \lambda_1).$$

*At a zero, necessarily simple,  $\nu = 1$ ,*

$$G'(\zeta) = C_\zeta \zeta^{-1}, \quad G''(\zeta) = C_\zeta \zeta^{-2} (2\lambda_1 - 1).$$

*Proof.* Substitute  $R = 0, 1, 2$  into (12.11), using  $s(2, 1) = -1$ ,  $s(2, 2) = 1$ , and  $\mathcal{B}_2(x_1, x_2) = x_1^2 + x_2$ .  $\square$

**Example 12.5** (A zero at  $-1$ ). Take  $G_{4,1}(q) = 1 + q + q^2 + q^3$ . At  $\zeta = -1$ ,

$$\nu = 1, \quad C_{-1} = -2, \quad \lambda_1 = \frac{3}{2}, \quad \lambda_2 = \frac{13}{12}.$$

Thus  $G'(-1) = 2$  and  $G''(-1) = -4$ , in agreement with

$$G_{4,1}(-1 + h) = 2h - 2h^2 + h^3.$$

At  $\zeta = 1$ ,  $C_1 = \binom{n}{k}$ ,  $\nu = 0$ , and  $\lambda_r = \kappa_r(n, k)$ : the expansion at one is recovered. Together, Chapter 10 and Theorems 11.5 and 12.3 give explicit coefficients at every complex base.

## Chapter 13

# Weighted sums and differentiated identities

### 13.1 Polynomial weights in the summation index

The finite product theorem can be differentiated in its auxiliary variable without differentiating the parameter  $q$ . This gives a second, independent source of weighted Gaussian sums. Let

$$P_N(z) = (-z; q)_N = \sum_{k=0}^N q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k, \quad \Theta_z = z \frac{d}{dz}.$$

**Theorem 13.1** (All power-weighted finite sums). *At a point where  $P_N(z) \neq 0$ , define*

$$K_r(z) = - \sum_{j=0}^{N-1} \text{Li}_{1-r}(-zq^j), \quad r \geq 1.$$

*Then, for  $r \geq 0$ ,*

$$\sum_{k=0}^N k^r q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k = P_N(z) \mathcal{B}_r(K_1(z), \dots, K_r(z)). \quad (13.1)$$

*The right side, after cancellation, is a polynomial and therefore extends to the zeros of  $P_N$ .*

*Proof.* The left side is  $\Theta_z^r P_N(z)$ . For every  $r \geq 1$ ,  $\Theta_z^r \log P_N(z) = K_r(z)$ . Taylor expansion of  $P_N(ze^t)/P_N(z)$  and the Bell generating function give the result. The polynomial extension follows because the two rational expressions agree away from finitely many zeros.  $\square$

For example,

$$\sum_{k=0}^N k q^{\binom{k}{2}} \begin{bmatrix} N \\ k \end{bmatrix}_q z^k = (-z; q)_N \sum_{j=0}^{N-1} \frac{zq^j}{1 + zq^j}.$$

For a general polynomial  $p(k) = \sum_{r=0}^R a_r k^r$ , sum (13.1) with weights  $a_r$ . Alternatively, expand  $p$  in falling factorials and use ordinary derivatives in  $z$ .

### 13.2 Alternating sums at a product zero

At  $z = 1$ , the product  $(z; q)_N$  vanishes for  $N \geq 1$ , so a logarithmic formula should first be deflated. Set

$$H_N(z) = (zq; q)_{N-1}, \quad (z; q)_N = (1 - z)H_N(z).$$

Assume for the moment that  $q^j \neq 1$  for  $1 \leq j < N$ , and put

$$J_s = -(s-1)! \sum_{j=1}^{N-1} \frac{q^{js}}{(1-q^j)^s} \quad (s \geq 1). \quad (13.2)$$

**Proposition 13.2** (Alternating factorial moments). *For  $r \geq 1$ ,*

$$\sum_{k=0}^N (-1)^k q^{\binom{k}{2}} \left[ \begin{matrix} N \\ k \end{matrix} \right]_q k^r = -r(q; q)_{N-1} \mathcal{B}_{r-1}(J_1, \dots, J_{r-1}). \quad (13.3)$$

For  $r = 0$ , the sum is zero. Ordinary powers are recovered by  $k^r = \sum_{a=0}^r S(r, a) k^{\underline{a}}$ .

*Proof.* The left side is the  $r$ -th derivative of  $(z; q)_N$  at one. Leibniz's rule and the factor  $1 - z$  leave only  $-r H_N^{(r-1)}(1)$ . The ordinary derivatives of  $\log H_N(z)$  at one are exactly  $J_s$ , and  $H_N(1) = (q; q)_{N-1}$ . Apply the ordinary Bell derivative formula, which follows from the Taylor expansion of  $H_N(1+t)/H_N(1)$ . The zeroth-order sum is  $(1; q)_N = 0$ . The conversion of powers to falling factorials is the defining ordinary second-kind Stirling expansion.  $\square$

When  $r > N$ , the left side is zero, and the right side is also zero: it represents a derivative of the degree- $N-1$  polynomial  $H_N$  of order  $r-1 > N-1$ . At excluded roots of unity, polynomial specialization or the root deflation of Theorem 12.3 replaces termwise evaluation of (13.2).

#### Which cancellation is actually present?

The Gaussian alternating kernel does not simply annihilate every polynomial weight of degree less than  $N$ . Already at  $N = 2$ ,

$$\sum_{k=0}^2 (-1)^k q^{\binom{k}{2}} \left[ \begin{matrix} 2 \\ k \end{matrix} \right]_q k = q - 1,$$

which is nonzero for generic  $q$ . What the product theorem gives is cancellation at specified *geometric spectral points*. The Jackson stencil and the extrapolation filters below include the necessary additional powers of  $q$ . Replacing every ordinary binomial coefficient in an additive finite-difference identity by a Gaussian coefficient is not a valid rule.

### 13.3 An all-order differentiation of central Vandermonde

The central convolution proved earlier is

$$G_{2n,n}(q) = \sum_{k=0}^n q^{k^2} G_{n,k}(q)^2.$$

Differentiating this identity at  $q = 1$  produces infinitely many finite identities involving ordinary binomial coefficients, Bernoulli numbers, and Bell polynomials. The logarithmic coordinate makes the formula uniform.

**Theorem 13.3** (Differentiated central convolution). *For  $r \geq 0$ , define, for each  $k$ ,*

$$v_1^{(k)} = nk, \quad v_a^{(k)} = 2\kappa_a(n, k) \quad (a \geq 2).$$

Then

$$\binom{2n}{n} \mathcal{B}_r(\kappa_1(2n, n), \dots, \kappa_r(2n, n)) = \sum_{k=0}^n \binom{n}{k}^2 \mathcal{B}_r(v_1^{(k)}, \dots, v_r^{(k)}). \quad (13.4)$$



*Proof.* Substitute  $q = e^t$ . On the  $k$ -th summand the logarithm, after removing the constant  $\binom{n}{k}^2$ , is

$$k^2 t + 2 \sum_{a \geq 1} \kappa_a(n, k) \frac{t^a}{a!}.$$

Its linear coefficient is  $k^2 + 2\kappa_1(n, k) = k^2 + k(n - k) = nk$ , and its remaining cumulants are  $2\kappa_a(n, k)$ . Expand both sides by Theorem 11.2 and compare coefficients of  $t^r/r!$ .  $\square$

For  $r = 1$  and  $n \geq 1$ , this gives

$$\sum_{k=0}^n k \binom{n}{k}^2 = \frac{n}{2} \binom{2n}{n}.$$

For  $r = 2$ , it gives the less immediate identity

$$\sum_{k=0}^n \binom{n}{k}^2 \left( n^2 k^2 + \frac{k(n-k)(n+1)}{6} \right) = \binom{2n}{n} \left( \frac{n^4}{4} + \frac{n^2(2n+1)}{12} \right).$$

Higher identities require no new differentiation argument: the same finite Bell formula gives their complete coefficients. The same procedure works for every block convolution in Chapter 3; each explicit power of  $q$  adds only to the first cumulant.

## Chapter 14

# Gaussian filters on geometric grids

### 14.1 Spectral design instead of recursive extrapolation

Let  $0 < \rho < 1$  and define the dilation operator  $(T_\rho f)(h) = f(\rho h)$ . Its eigenfunctions are the monomials:  $T_\rho h^m = \rho^m h^m$ . To preserve a constant and annihilate the first  $N$  positive powers, construct the normalized spectral polynomial

$$R_N = \prod_{a=1}^N \frac{T_\rho - \rho^a}{1 - \rho^a}. \quad (14.1)$$

Here  $R_0$  is the identity operator. This is a geometric version of the source's principle of choosing a basis in which an operator is diagonal.

**Theorem 14.1** (Closed Gaussian extrapolation weights). *For every  $N \geq 0$ ,*

$$(R_N f)(h) = \sum_{j=0}^N w_{N,j} f(\rho^j h), \quad (14.2)$$

$$w_{N,j} = \frac{(-1)^{N-j} \rho^{\binom{N-j+1}{2}} \left[ \begin{matrix} N \\ j \end{matrix} \right]_\rho}{(\rho; \rho)_N}. \quad (14.3)$$

*The filter preserves constants and annihilates  $h^m$  for  $1 \leq m \leq N$ . For every integer  $m \geq N+1$ , its complete remaining response is*

$$R_N h^m = (-1)^N \rho^{\binom{N+1}{2}} \left[ \begin{matrix} m-1 \\ N \end{matrix} \right]_\rho h^m. \quad (14.4)$$

*Proof.* Expand the numerator polynomial in (14.1). The coefficient of  $T_\rho^j$  is  $(-1)^{N-j} e_{N-j}(\rho, \dots, \rho^N)$ . Principal specialization of the elementary symmetric function gives

$$e_{N-j}(\rho, \dots, \rho^N) = \rho^{\binom{N-j+1}{2}} \left[ \begin{matrix} N \\ N-j \end{matrix} \right]_\rho,$$

and Gaussian symmetry gives (14.3). Evaluate the spectral polynomial at  $1, \rho, \dots, \rho^N$  to prove preservation and annihilation. For  $m \geq N+1$ , direct factorization gives

$$\prod_{a=1}^N \frac{\rho^m - \rho^a}{1 - \rho^a} = (-1)^N \rho^{N(N+1)/2} \prod_{a=1}^N \frac{1 - \rho^{m-a}}{1 - \rho^a} = (-1)^N \rho^{\binom{N+1}{2}} \left[ \begin{matrix} m-1 \\ N \end{matrix} \right]_\rho.$$

This is (14.4). □

**Corollary 14.2** (Exact analytic error series and a bound). *Suppose  $f(h) = L + \sum_{m \geq 1} c_m h^m$  is analytic for  $|h| < R$ . Then, throughout that disk,*

$$(R_N f)(h) - L = (-1)^N \rho^{\binom{N+1}{2}} \sum_{m \geq N+1} \left[ \begin{matrix} m-1 \\ N \end{matrix} \right]_{\rho} c_m h^m. \quad (14.5)$$

*If, additionally,  $f$  is analytic on a neighborhood of the closed disk  $|h| \leq R$  and  $|f(h)| \leq M$  there, then, for  $x = |h|/R < 1$ ,*

$$|(R_N f)(h) - L| \leq M \rho^{\binom{N+1}{2}} \frac{x^{N+1}}{(x; \rho)_{N+1}}. \quad (14.6)$$

*Proof.* The operator is a finite combination of evaluations at points inside the disk, so it can be applied term by term to the convergent power series. Use (14.4). Cauchy's coefficient bound gives  $|c_m| \leq MR^{-m}$ . All Gaussian values are nonnegative at  $0 < \rho < 1$ , and the reciprocal finite-product theorem gives

$$\sum_{m \geq N+1} \left[ \begin{matrix} m-1 \\ N \end{matrix} \right]_{\rho} x^m = \frac{x^{N+1}}{(x; \rho)_{N+1}}.$$

Taking absolute values therefore proves (14.6).  $\square$

## 14.2 Noise amplification is bounded at a fixed geometric ratio

**Theorem 14.3** (Exact absolute-weight norm). *For  $0 < \rho < 1$ ,*

$$\sum_{j=0}^N |w_{N,j}| = \frac{(-\rho; \rho)_N}{(\rho; \rho)_N} \leq \prod_{a \geq 1} \frac{1 + \rho^a}{1 - \rho^a} < \infty. \quad (14.7)$$

*Thus the bound is uniform in  $N$  when  $\rho$  is fixed. One explicit, though not sharp, estimate is*

$$\sum_{j=0}^N |w_{N,j}| \leq \exp \left( \frac{2\rho}{(1 - \rho)^2} \right). \quad (14.8)$$

*If each sample has absolute error at most  $\varepsilon$ , the resulting filtered error is at most  $\varepsilon$  times (14.7).*

*Proof.* After taking absolute values, the numerator sum in (14.3) is

$$\sum_{b=0}^N \rho^{\binom{b+1}{2}} \left[ \begin{matrix} N \\ b \end{matrix} \right]_{\rho} = (-\rho; \rho)_N$$

by the finite product theorem. For  $0 \leq u < 1$ ,

$$\log \frac{1+u}{1-u} = 2 \sum_{r \geq 0} \frac{u^{2r+1}}{2r+1} \leq \frac{2u}{1-u}.$$

Consequently the logarithm of the infinite product is at most  $\sum_{a \geq 1} 2\rho^a / (1 - \rho^a) \leq 2\rho / (1 - \rho)^2$ , which proves convergence and the bound. The final assertion is the triangle inequality applied to (14.2).  $\square$

The fixed-ratio hypothesis matters. The bound need not remain moderate as  $\rho$  approaches one. Nor does it control the error in generating the samples themselves; the theorem assumes a bound on those errors and describes only the linear combination's amplification.

**Example 14.4** (A dyadic three-point formula). For  $\rho = 1/2$  and  $N = 2$ ,

$$(R_2 f)(h) = \frac{1}{3}f(h) - 2f(h/2) + \frac{8}{3}f(h/4). \quad (14.9)$$

It preserves constants, cancels the linear and quadratic terms, and has leading error  $c_3 h^3/8$ . Its exact absolute-weight norm is five. More generally, (14.5) specifies *every* remaining power, not just the leading one.

### 14.3 Fractional powers and arithmetic exponent lattices

For positive  $h$ , let the unwanted exponents be  $\alpha, \alpha + \delta, \dots, \alpha + (N-1)\delta$ , where  $\alpha, \delta > 0$ . Define

$$R_{N;\alpha,\delta} = \prod_{a=0}^{N-1} \frac{T_\rho - \rho^{\alpha+a\delta}}{1 - \rho^{\alpha+a\delta}}. \quad (14.10)$$

It preserves constants and annihilates exactly those prescribed power modes. The same elementary-symmetric specialization gives the closed weights

$$[T_\rho^j] R_{N;\alpha,\delta} = \frac{(-1)^{N-j} \rho^{\alpha(N-j)+\delta \binom{N-j}{2}} \begin{bmatrix} N \\ j \end{bmatrix}_{\rho^\delta}}{(\rho^\alpha; \rho^\delta)_N}. \quad (14.11)$$

This is proved exactly as (14.3), now with the alphabet  $\rho^\alpha, \rho^{\alpha+\delta}, \dots$ . For a general exponent  $\beta$ , the full response is the explicit product

$$R_{N;\alpha,\delta} h^\beta = h^\beta \prod_{a=0}^{N-1} \frac{\rho^\beta - \rho^{\alpha+a\delta}}{1 - \rho^{\alpha+a\delta}}.$$

In particular, an even-power error expansion in the original step size uses  $\alpha = \delta = 2$ , and hence the Gaussian base is  $\rho^2$ , not  $\rho$ .

### 14.4 Logarithmic errors and repeated spectral roots

The geometric operator also handles polynomial-logarithmic errors. The key identity is

$$p(T_\rho)(h^\beta (\log h)^r) = \frac{\partial^r}{\partial \beta^r} (h^\beta p(\rho^\beta)), \quad h > 0. \quad (14.12)$$

Indeed,  $h^\beta (\log h)^r = \partial_\beta^r h^\beta$ , and  $p(T_\rho)$  is a finite sum of substitutions independent of  $\beta$ .

**Proposition 14.5** (Confluent geometric cancellation). *If  $p(z)$  has a zero of multiplicity at least  $s$  at  $z = \rho^\beta$ , then  $p(T_\rho)$  annihilates  $h^\beta (\log h)^r$  for every  $0 \leq r < s$ . For distinct positive exponents  $\beta_1, \dots, \beta_A$  and positive integers  $s_1, \dots, s_A$ , the normalized filter*

$$p(z) = \prod_{a=1}^A \left( \frac{z - \lambda_a}{1 - \lambda_a} \right)^{s_a}, \quad \lambda_a = \rho^{\beta_a}, \quad (14.13)$$

*preserves constants and cancels all the corresponding logarithmic modes. Its coefficients have the recurrence-free Bell formula*

$$C_0 = \prod_{a=1}^A \frac{(-\lambda_a)^{s_a}}{(1 - \lambda_a)^{s_a}}, \quad (14.14)$$

$$[z^\ell] p(z) = \frac{C_0}{\ell!} \mathcal{B}_\ell \left( -0! \sum_a s_a \lambda_a^{-1}, -1! \sum_a s_a \lambda_a^{-2}, \dots, -(\ell-1)! \sum_a s_a \lambda_a^{-\ell} \right). \quad (14.15)$$

For  $\ell > \sum_a s_a$ , the latter expression is identically zero.

*Proof.* The derivative of  $\rho^\beta$  with respect to  $\beta$  is  $(\log \rho)\rho^\beta \neq 0$ . A zero of order  $s$  in the spectral coordinate therefore remains a zero of order  $s$  in the exponent coordinate. Applying (14.12) with  $r < s$  gives zero. Formula (14.13) has all these zeros and satisfies  $p(1) = 1$ .

For its coefficients, write

$$p(z) = C_0 \prod_{a=1}^A (1 - z/\lambda_a)^{s_a}, \quad \log \frac{p(z)}{C_0} = - \sum_{r \geq 1} \left( \sum_a s_a \lambda_a^{-r} \right) \frac{z^r}{r}.$$

Exponentiation is exactly the Bell master formula. The degree assertion follows because  $p$  is a polynomial of degree  $\sum_a s_a$ .  $\square$

For an asymptotic expansion rather than a convergent series, the cancellation claims apply to any finite collection of terms for which a remainder estimate has been established. A finite filter transports that remainder by the triangle inequality. One should not apply an infinite formal expansion as an equality of functions without a separate convergence or asymptotic argument.

## 14.5 Connection with the Fabius-function setting

The relevant bridge to the surrounding repository is the geometric grid and its spectral algebra. Whenever an approximation to a Fabius- or Rvachev-related quantity is known to have an expansion in powers of a dyadic scale, its known error terms can be canceled by the corresponding filter. For a grid ratio of  $1/2$ , integral powers use Gaussian base  $1/2$ ; an even-power expansion uses base  $1/4$ . Logarithmic corrections require repeated roots, as just proved.

This application is deliberately conditional on the expansion of the particular approximation being used. No assertion here identifies a specific Fourier normalization, spline sequence, or exact dyadic Fabius value with such an error series without proving that additional bridge. The unconditional result is the filter algebra, including explicit weights, complete power response, analytic remainder bound, and noise norm. That separation makes the result reusable without importing hidden normalization assumptions.

## Chapter 15

# Multinomial coefficients and factorial ratios

### 15.1 The Gaussian multinomial extension

For  $k_1 + \cdots + k_s = n$ , define

$$G_{k_1, \dots, k_s}(q) = \frac{(q; q)_n}{\prod_{i=1}^s (q; q)_{k_i}} = \begin{bmatrix} n \\ k_1 \end{bmatrix}_q \begin{bmatrix} n - k_1 \\ k_2 \end{bmatrix}_q \cdots \begin{bmatrix} n - k_1 - \cdots - k_{s-1} \\ k_s \end{bmatrix}_q.$$

The product expression proves polynomiality, nonnegative integer coefficients, and the value  $n! / \prod_i k_i!$  at one. The degree is

$$D = \sum_{i < j} k_i k_j = \frac{1}{2} \left( n^2 - \sum_i k_i^2 \right).$$

The last expression follows by adding the degrees in the product; reciprocity follows by multiplying the Gaussian reciprocity identities.

**Proposition 15.1** (Multinomial coefficient and parameter kernels). *For  $r \geq 1$ , put*

$$A_r = \sum_{i=1}^s \sum_{\substack{d|r \\ d \leq k_i}} d - \sum_{\substack{d|r \\ d \leq n}} d.$$

*Then*

$$[q^R] G_{k_1, \dots, k_s}(q) = \frac{1}{R!} \mathcal{B}_R(0!A_1, 1!A_2, \dots, (R-1)!A_R). \quad (15.1)$$

*Its logarithmic-parameter cumulants at one are*

$$\kappa_1 = \frac{D}{2}, \quad \kappa_r = \frac{\beta_r}{r} \left( P_r(n) - \sum_{i=1}^s P_r(k_i) \right) \quad (r \geq 2), \quad (15.2)$$

*where  $P_r(M) = \sum_{j=1}^M j^r$ . Its multiplicity at a primitive  $d$ -th root is*

$$\nu_d = \left\lfloor \frac{n}{d} \right\rfloor - \sum_{i=1}^s \left\lfloor \frac{k_i}{d} \right\rfloor \in \{0, 1, \dots, s-1\}. \quad (15.3)$$

*Proof.* For the coefficient formula, take the formal logarithm at zero. Each  $\log(1 - q^j)$  contributes  $-j/r$  to the coefficient of  $q^r$  when  $j \mid r$ , and zero otherwise. Subtracting denominator logarithms gives  $A_r/r$ , and Bell expansion proves (15.1). The calculation at one is the factor-by-factor regular logarithm of Theorem 11.1, with  $P_r(n) - \sum_i P_r(k_i)$  in place of the two-block power-sum difference. For the multiplicity, count multiples of  $d$  in the factorials. Writing  $k_i = a_i d + b_i$ ,  $0 \leq b_i < d$ , gives  $\nu_d = \lfloor \sum_i b_i/d \rfloor$ , proving the stated range.  $\square$

Unlike a Gaussian binomial polynomial, a Gaussian multinomial need not be squarefree. For example, the polynomial  $G_{1,1,1,1}(q) = [4]_q!$  has multiplicity two at  $-1$ :

$$G_{1,1,1,1}(q) = (1+q)^2(1+q+q^2)(1+q^2). \quad (15.4)$$

For its local jets, define  $C_\zeta$  and  $\lambda_r$  by taking all exponents  $1, \dots, n$  in the numerator and all exponents  $1, \dots, k_i$ , with multiplicities, in the denominator of (12.7) and (12.8). Then (12.9) and (12.11) hold verbatim with  $\nu = \nu_d$ . Their proof was factorwise and never required  $\nu \leq 1$ . In this way the same Bell formula handles higher-order zeros without a new inversion calculation.

### 15.1.1 A multinomial Lucas formula

Let  $n = ad + b$ ,  $k_i = a_i d + b_i$ , with all remainders in  $\{0, \dots, d-1\}$ . If  $\sum_i b_i > b$ , then  $\nu_d > 0$  and the multinomial vanishes at a primitive  $d$ -th root. If  $\sum_i b_i = b$ , there is no carry and

$$G_{k_1, \dots, k_s}(\zeta) = \binom{a}{a_1, \dots, a_s} G_{b_1, \dots, b_s}(\zeta). \quad (15.5)$$

Indeed, use the product of Gaussian binomials and apply  $q$ -Lucas at each step. Without a carry, the ordinary binomial factors multiply to the ordinary multinomial, and the remainder factors multiply to the smaller Gaussian multinomial. With a carry, the multiplicity calculation already proves vanishing.

## 15.2 A polynomiality criterion for balanced factorial ratios

Let  $a_1, \dots, a_u$  and  $b_1, \dots, b_v$  be nonnegative integers with  $\sum_i a_i = \sum_j b_j$ , and set

$$F(q) = \frac{\prod_{i=1}^u (q; q)_{a_i}}{\prod_{j=1}^v (q; q)_{b_j}}.$$

Let  $M$  be the largest index, taking  $M = 0$  if every index is zero.

**Theorem 15.2** (Finite cyclotomic criterion). *The rational function  $F(q)$  is a polynomial if and only if*

$$E_d := \sum_i \left\lfloor \frac{a_i}{d} \right\rfloor - \sum_j \left\lfloor \frac{b_j}{d} \right\rfloor \geq 0 \quad (2 \leq d \leq M). \quad (15.6)$$

When these conditions hold,

$$F(q) = \prod_{d=2}^M \Phi_d(q)^{E_d} \in \mathbb{Z}[q], \quad F(0) = 1. \quad (15.7)$$

Its degree and value at one are

$$D = \frac{1}{2} \left( \sum_i a_i^2 - \sum_j b_j^2 \right), \quad F(1) = \frac{\prod_i a_i!}{\prod_j b_j!}. \quad (15.8)$$

It satisfies  $F(q^{-1}) = q^{-D} F(q)$ . All its parameter jets follow from the same factorwise deflation as Theorem 12.3.

*Proof.* Cyclotomic factorization gives  $F = \prod_{d=1}^M \Phi_d^{E_d}$ . The balancing hypothesis makes  $E_1 = 0$  and cancels the signs from  $1 - q^a$ . Distinct cyclotomic factors have disjoint roots. A negative  $E_d$  would therefore give a genuine pole and prevent polynomiality; nonnegative exponents give the integer polynomial (15.7). The degree is the difference of the sums  $1 + \cdots + a_i$  and  $1 + \cdots + b_j$ ; balancing cancels their linear parts. Cancelling  $1 - q$  from every individual factor before taking  $q \rightarrow 1$  gives the factorial value. Finally,  $1 - q^{-r} = -q^{-r}(1 - q^r)$ , so the same degree and sign calculation gives reciprocity. The proof of Theorem 12.3 applies to this multiset of factors without change.  $\square$

Polynomiality does not imply coefficientwise positivity. A small balanced example is

$$\frac{(q; q)_6 (q; q)_1^2}{(q; q)_5 (q; q)_3} = \frac{(1 - q^6)(1 - q)}{(1 - q^3)(1 - q^2)} = 1 - q + q^2 = \Phi_6(q).$$

Thus the finite-field and partition interpretations of Gaussian coefficients supply information beyond the cyclotomic criterion. Conversely, the criterion extends the local-series machinery to examples for which no positive counting interpretation is available.



## Chapter 16

# Computation, verification, and integration

### 16.1 Choosing a representation for the task

The formulae above are equal algebraically, but they serve different computational purposes. A recurrence-free expression is a useful mathematical specification; it is not automatically the fastest or best-conditioned numerical algorithm.

For an exact Gaussian polynomial, the division-free Pascal recurrence is usually a natural baseline: it uses only integer polynomial addition and shifts. For a single coefficient in the base, Chapter 10 supplies a finite divisor-sum Bell expression or a restricted-partition expression. For a coefficient of a product in an auxiliary variable, the geometric-product master formula avoids expanding all factors separately. One first computes a finite list of power sums and then evaluates one Bell polynomial.

For symbolic derivatives at a generic parameter, the logarithmic derivatives  $L_r$  in Theorem 11.5 compress large product-rule expressions. At one, Bernoulli cumulants are simpler. At another root of unity, use the constant, multiplicity, and regular logarithmic data  $(C_\zeta, \nu, \lambda_r)$  from Theorem 12.3. Do not use a product quotient as a numerical  $0/0$  and hope that floating-point cancellation will reconstruct the limit.

For basis conversion, the triangular  $q$ -Stirling recurrences give a convenient algorithm, while the finite Gaussian sum and Bell diagonal formulae give independent specifications. For implicit series, compute the logarithmic power sums of the defining product and apply Theorem 9.2; the ordinary Lagrange factor remains  $1/n$ . For extrapolation, compute the weights directly from (14.3) and use the error and noise formulae together, rather than increasing the cancellation order without controlling sample accuracy.

### 16.2 Arithmetic cost and finite verification

Suppose the power sums  $P_1, \dots, P_R$  are available. The coefficients  $f_0, \dots, f_R$  of  $F(z) = \exp(\sum_{r \geq 1} P_r z^r / r)$  can be evaluated through

$$f_0 = 1, \quad n f_n = \sum_{r=1}^n P_r f_{n-r}. \quad (16.1)$$

This uses  $O(R^2)$  arithmetic operations and  $O(R)$  stored scalars. It follows by differentiating  $F$  and comparing coefficients in  $zF'(z) = F(z) \sum_{r \geq 1} P_r z^r$ . The algorithm is recurrent, but its output is specified independently by the recurrence-free Bell sum (4.9). The distinction permits efficient calculation and an independent formula audit at the same time. Bit complexity also depends on coefficient heights and is not described by the arithmetic-operation count.

The companion `verify_identities.py` uses only Python’s standard library, integer polynomial arithmetic, and exact rational numbers. Its successful run for this article reports

**10,912 exact checks in 36 groups.**

The tests include Gaussian products and convolution, both terminating  $q$ -Chu–Vandermonde sums, divisor identities, Jackson stencils and Taylor reconstruction,  $q$ -Stirling sums and Bell diagonals, weighted composition, Lagrange coefficients, cyclotomic factors, parameter cumulants, all-root local jets, and ordinary and logarithmic extrapolation filters.

The checks deliberately use different representations. Gaussian polynomials are constructed by polynomial Pascal addition and compared with product values; Bell expressions are checked against direct finite products or moments; Lagrange coefficients are compared with ordinary truncated fixed-point substitution; and  $q$ -Lucas is checked as exact divisibility by a cyclotomic polynomial. Root-of-unity jets are checked not just at  $-1$ , but in the exact fields  $\mathbb{Q}(\zeta_d)$ ,  $3 \leq d \leq 8$ , using polynomials modulo  $\Phi_d$ . For  $u$  of exact order  $L > 1$ , the finite certificate

$$(1 - u)^{-1} = -\frac{1}{L} \sum_{j=1}^{L-1} j u^j$$

provides exact division in these calculations: multiplication by  $1 - u$  reduces the numerator sum to  $-L$ .

Polynomial coefficient and moment tests cover  $0 \leq n \leq 12$ , all  $0 \leq k \leq n$ , and orders through eight. The  $-1$  tests cover orders through nine. The cyclotomic-field jet tests cover  $0 \leq n \leq 9$ , all lower indices, and orders through seven. Other test bounds and the exact rational parameter values are specified directly in the program. The archive includes the complete output as `validation_report.txt`.

#### What this verification establishes

These deterministic finite checks are reproducible audits of signs, shifts, normalizations, and coefficient formulae. They are not proofs for unbounded indices, do not establish floating-point stability outside the stated bounds, and do not assert Lean verification. The mathematical proofs in the preceding chapters establish the general statements. No current declaration inventory or formalization status of the linked repository is inferred from these tests.

### 16.3 A theorem-level insertion map

The following map identifies the mathematical interface with the source without assuming that its chapter numbering will remain fixed.

| Source mechanism                           | Extension supplied here                                                                                                                  |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Newton expansion and factorial bases       | Geometric Jackson–Newton interpolation, explicit Gaussian stencils, and the $[j]_q$ -node $q$ -Stirling arrays (Chapters 6 and 7).       |
| Exponential formula and Bell polynomials   | Finite-alphabet power sums, arbitrary geometric-product exponents, determinants, and inverse logarithmic coordinates (Chapter 4).        |
| Binomial inversion and incidence algebra   | Gaussian inversion, subspace-lattice Möbius coefficients, and positive linearization (Chapter 3).                                        |
| Stirling, Bernoulli, and Eulerian families | Parameter derivatives and cumulants, ordinary/logarithmic derivative conversion, and regular root-of-unity factors (Chapters 11 and 12). |

| Source mechanism                    | Extension supplied here                                                                                                             |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Riordan and composition calculus    | A specified $q$ -factorial normalization of ordinary composition and its matrix group law (Chapter 8).                              |
| Lagrange–Bürmann and Good inversion | Product-defined inverses with finite Bell coefficients and a fully computed cross-coupled two-variable case (Chapter 9).            |
| Asymptotic coefficient operators    | Geometric spectral filters, complete error coefficients, noise norms, and confluent cancellation of logarithmic modes (Chapter 14). |

## 16.4 Concluding perspective

Gaussian coefficients occur here in three complementary roles. They are coefficients of elementary and complete symmetric functions on geometric alphabets; they are kernels for triangular basis changes and inversions; and they are the explicit weights of spectral polynomials on geometric grids. Ordinary Bell polynomials connect these roles by turning logarithmic power sums into finite coefficients. Stirling numbers change the derivative coordinate, Bernoulli numbers remove the apparent singularity at one, and Eulerian polynomials make the remaining logarithmic derivatives rational.

This common mechanism is more informative than a formal replacement of every integer by a  $q$ -integer. It explains which structures survive unchanged, which acquire a Gaussian kernel, and which require a different operation. It also separates exact finite algebra from local analysis and asymptotic remainder estimates. Those separations are what allow the extension to serve as a reusable part of the source’s coefficient calculus.

# Appendix A

## A compact formula atlas

This atlas records the input and output of the principal coefficient mechanisms. All symbols and domains are defined in the referenced results.

### Finite geometric products

For  $F(z) = \prod_{\ell} (a_{\ell} z; q^{d_{\ell}})_{N_{\ell}}^{-\alpha_{\ell}}$ , put  $P_r = \sum_{\ell} \alpha_{\ell} a_{\ell}^r [N_{\ell}]_{q^{d_{\ell} r}}$ . Then

$$[z^n]F(z) = \frac{1}{n!} \mathcal{B}_n(((r-1)!P_r)_{r=1}^n) = \sum_{\sum r m_r = n} \prod_{r=1}^n \frac{P_r^{m_r}}{r^{m_r} m_r!}.$$

This is Theorem 4.2. For an implicit solution  $y = zF(y)$ , replace  $P_r$  by  $nP_r$ , use Bell degree  $n-1$ , and divide by  $n!$ , as in Theorem 9.2.

### Gaussian polynomial coefficients

For  $m = n - k$ , set

$$A_r = \sum_{\substack{d|r \\ 1 \leq d \leq k}} d - \sum_{\substack{d|r \\ m < d \leq n}} d.$$

Then

$$[q^R] \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{1}{R!} \mathcal{B}_R(((r-1)!A_r)_{r=1}^R).$$

The two-block divisor kernels are the multiset difference of denominator and numerator exponents; see Chapter 10.

### The $q = 1$ expansion

Let  $\kappa_1 = k(n-k)/2$  and  $\kappa_r = (\beta_r/r) \sum_{j=1}^k ((n-k+j)^r - j^r)$ ,  $r \geq 2$ . Then

$$\begin{bmatrix} n \\ k \end{bmatrix}_{e^t} = \binom{n}{k} \sum_{r \geq 0} \mathcal{B}_r(\kappa_1, \dots, \kappa_r) \frac{t^r}{r!}.$$

For ordinary  $q$ -derivatives, apply signed Stirling coefficients as in (11.7).

## All roots of unity

Deflate every factor  $1 - \zeta^a e^{at}$ , using  $-at$  when  $\zeta^a = 1$  and  $1 - \zeta^a$  otherwise. The constant product is  $C_\zeta$ , the net order is  $\nu$ , and the regular logarithmic derivatives are  $\lambda_r$ . Then

$$\left[ \begin{matrix} n \\ k \end{matrix} \right]_{\zeta e^t} = C_\zeta t^\nu \sum_{r \geq 0} \mathcal{B}_r(\lambda_1, \dots, \lambda_r) \frac{t^r}{r!}.$$

Equations (12.5)–(12.8) supply every input by finite sums, and (12.11) converts to ordinary Taylor coefficients.

## Geometric extrapolation

For  $0 < \rho < 1$ , the weights

$$w_{N,j} = \frac{(-1)^{N-j} \rho^{\binom{N-j+1}{2}} \left[ \begin{matrix} N \\ j \end{matrix} \right]_\rho}{(\rho; \rho)_N}$$

preserve constants, kill the first  $N$  powers, and have exact absolute norm  $(-\rho; \rho)_N / (\rho; \rho)_N$ . Their response to every higher power is  $(-1)^N \rho^{\binom{N+1}{2}} \left[ \begin{matrix} m-1 \\ N \end{matrix} \right]_\rho$ . Repeated spectral roots cancel logarithmic modes; see Theorem 14.5.

## Appendix B

# Source provenance and reproducibility

### B.1 Source and scope

The requested source is the canonical *Combinatorial Coefficient Calculus* under the repository path given in [1]. The directory metadata read for this extension identified its  $\LaTeX$  file by Git blob SHA

ffbe7c117db3fbeb3c0ae804e84422bb2dc6a89b.

This is a content identifier for that file, not a claim about the repository’s latest commit or its formalization coverage. The source interface used here is its coefficient rules, Newton basis, Bell generating functions, Stirling normalizations, and composition/inversion organization. The present document is not a line-by-line revision of the source, and the original files were not modified.

The classical identities cited in the bibliography are proved in the text in the normalizations actually used. The root-jet and filter developments are presented with complete derivations, rather than a priority claim. No inference of novelty is made from the absence of a result in a small set of references.

### B.2 Files and commands

The archive contains the standalone  $\LaTeX$  source, its compiled PDF, `verify_identities.py`, `validation_report.txt`, and a `README.md`. No bibliography processor or repository-specific notation file is needed. With a  $\TeX$  installation containing Libertinus and the listed standard packages, compile with

```
pdflatex -interaction=nonstopmode -halt-on-error \  
  q_binomial_coefficient_calculus.tex  
pdflatex -interaction=nonstopmode -halt-on-error \  
  q_binomial_coefficient_calculus.tex
```

A third pass may be needed after substantial edits to the table of contents. Equivalently, use `latexmk -pdf q_binomial_coefficient_calculus.tex`. The verifier requires Python 3.9 or later and no third-party packages:

```
python verify_identities.py
```

The supplied report records the successful run accompanying this PDF. Running the program again recomputes the certificates; it does not read or trust the saved report.

# Bibliography

- [1] Vladimir Reshetnikov, *ProveIt repository: Combinatorial Coefficient Calculus*, canonical L<sup>A</sup>T<sub>E</sub>X monograph and accompanying provenance files. Source consulted September 4, 2026. [https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial\\_Coefficient\\_Calculus](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/combinatorial-coefficient-calculus/Combinatorial_Coefficient_Calculus).
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